

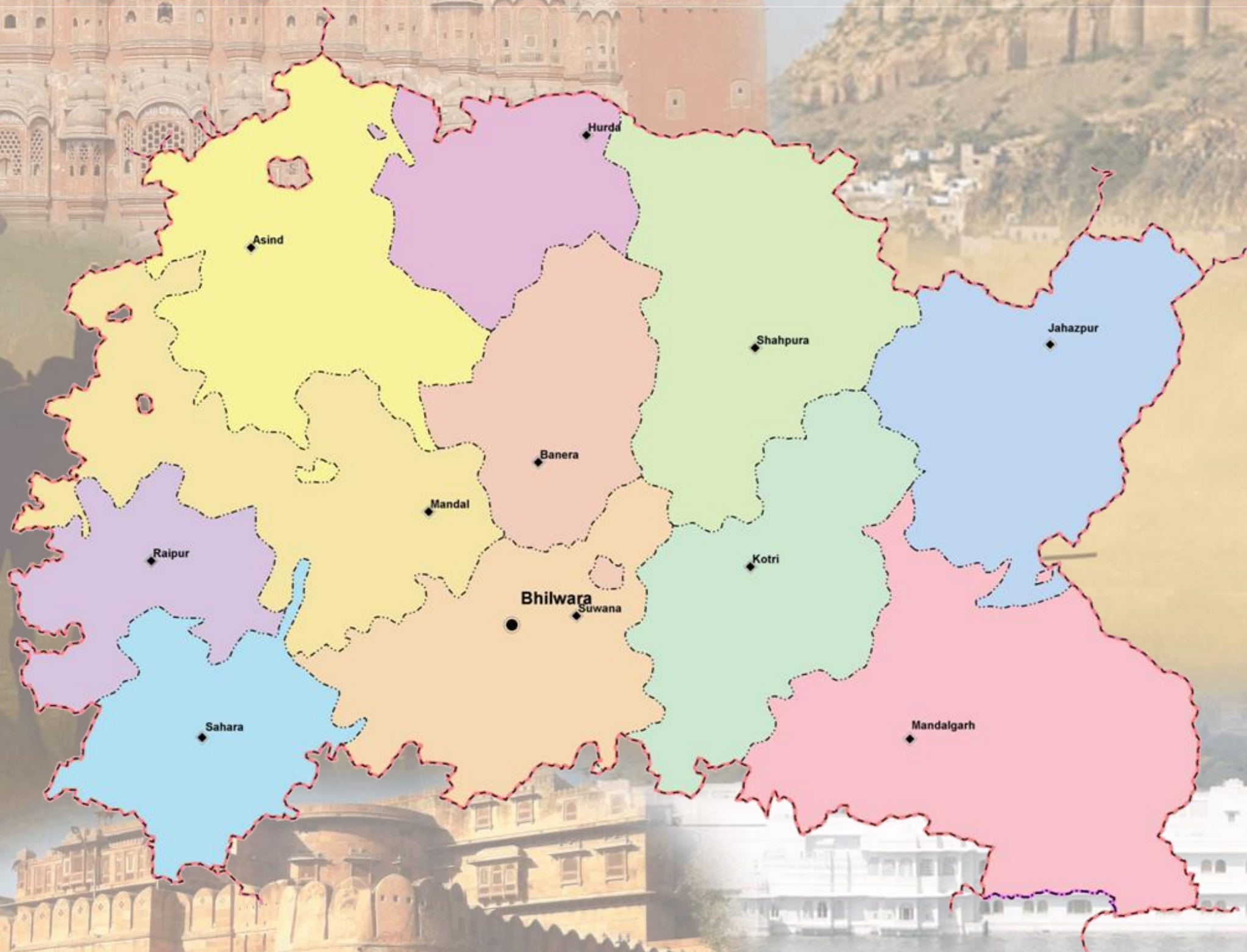


Ground Water Department,
Rajasthan

Hydrogeological Atlas of Rajasthan Bhilwara District



European Union
State Partnership Programme



2013



Rolta India Limited

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

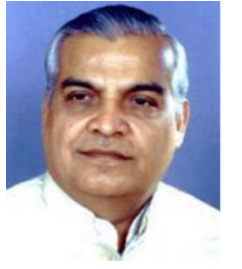
It is also heartening to note that the outcome of the project is being published in the form of 'Hydrogeological Atlas of Rajasthan'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.



(Ashok Gehlot)

Message

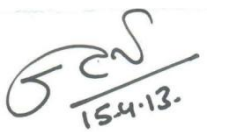


Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.



(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of 'Hydrogeological Atlas of Rajasthan' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Bhilwara District



2013

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ADMINISTRATIVE SETUP

DISTRICT – BHILWARA

Location:

Bhilwara district is located in the southern part of Rajasthan. It is bounded in the north by Ajmer and Tonk districts, in the east by Bundi district, south by Chittaurgarh district and the state of Madhya Pradesh and by Rajsamand district in the west. It stretches between 25° 00' 38.87" to 25° 57' 53.70" north latitude and 74° 00' 31.67" to 75° 27' 46.25" east longitude covering area of 10,445.1 sq km. Major part of the district has a systematic drainage system contributing to 'Banas River Basin' (central and western parts of the district) for its most part and the eastern part drains into the 'Chambal River Basin'.

Administrative Set-up:

Bhilwara district is administratively divided into eleven Blocks. The following table summarizes the basic statistics of the district at block level.

S. No.	Block Name	Population (Based on 2001 census)	Area (sq km)	% of District Area	Total Number of Towns and Villages
1	Asind	2,06,469	1,119.7	10.7	205
2	Banera	1,06,222	689.0	6.6	85
3	Hurda	1,15,450	615.9	5.9	73
4	Jahazpur	1,86,597	1,087.3	10.4	231
5	Kotri	1,48,702	938.7	9.0	175
6	Mandal	1,97,500	1,134.4	10.9	191
7	Mandalgarh	2,32,493	1,534.0	14.6	301
8	Raipur	84,354	542.1	5.2	92
9	Sahara	1,15,123	685.0	6.6	105
10	Shahpura	1,76,747	1,150.4	11.0	156
11	Suwana	4,44,132	948.6	9.1	139
Total		20,13,789	10,445.1	100.0	1,753

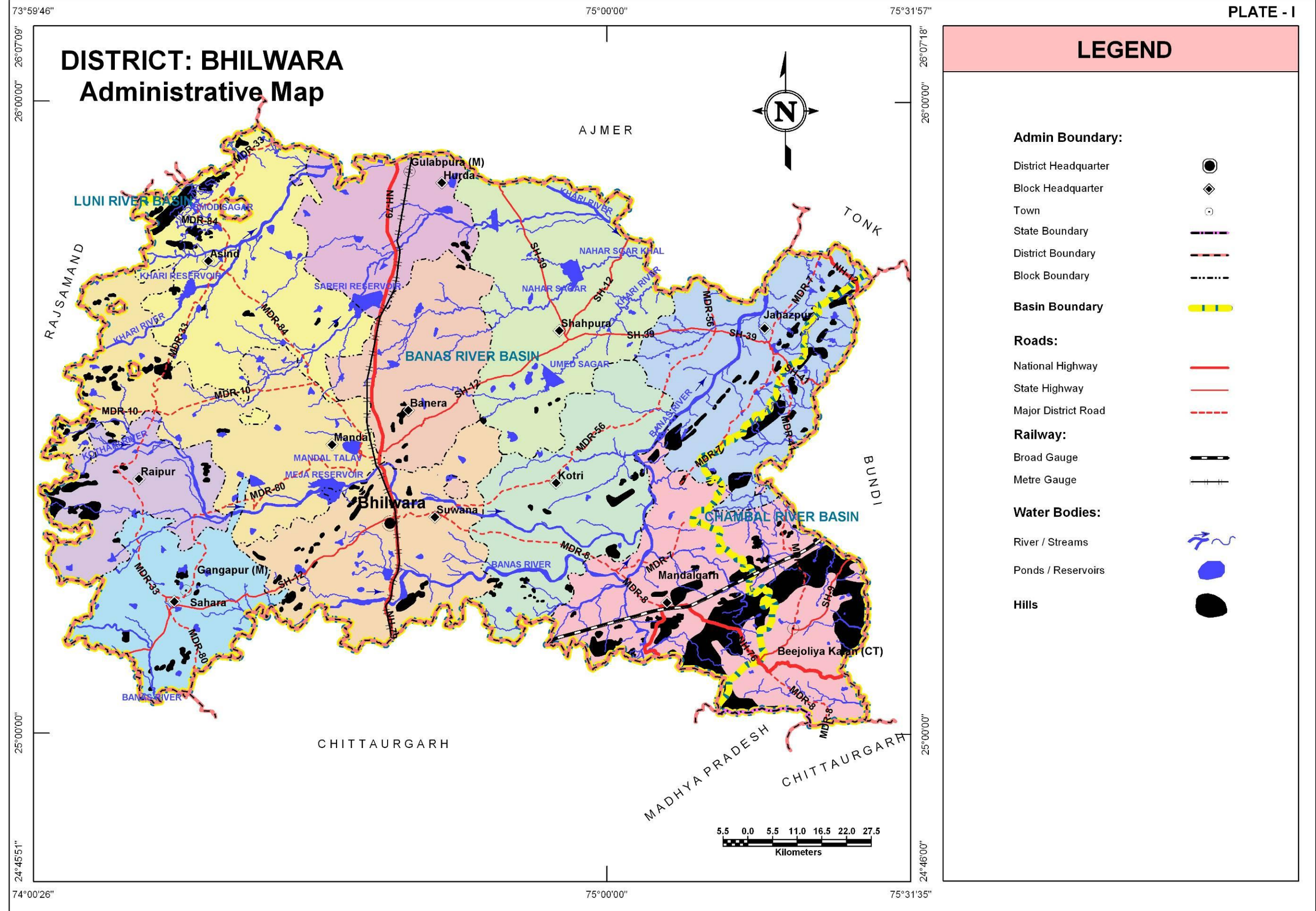
Bhilwara district has 1,753 towns and villages, out of which eleven are block headquarters as well.

Climate:

January is the coldest month with mean and maximum temperatures being lowest at 22.2 °C and 7.3 °C. Temperature during peak summer month of June reaches up to 46 °C. Monsoon months from July to September bring in rains to the district which help in not only contain the rising temperatures but also to replenish the water resources both surface and sub-surface. Atmosphere is generally dry except during the monsoon period. Annual mean rainfall in the district is 583.7mm.



PLATE - I



TOPOGRAPHY

The district is rectangular in shape except for its western portion which is comparatively broader than the eastern one. The topography of Bhilwara district is undulating plain. Hills in the southeastern and western parts follow a general northeast to southwest trend. Banas is the most important river which flows a west to east trend in the initial part of flow and then takes a SW to NE trend before leaving the district. There are some important tributaries like Kothari and Khari etc. Major reservoirs are Sareri and Khari in the northwest, Meja in the southwestern part of the district. The general topographic elevation in the district is between 375 m to 500 m above mean sea level. Elevation ranges from a minimum of 304.8m above mean sea level in Jahazpur block in the northeastern part of the district and maximum of 821.9 m above mean sea level In Asind block in northwestern part of the district.

Table: Block wise minimum and maximum elevation

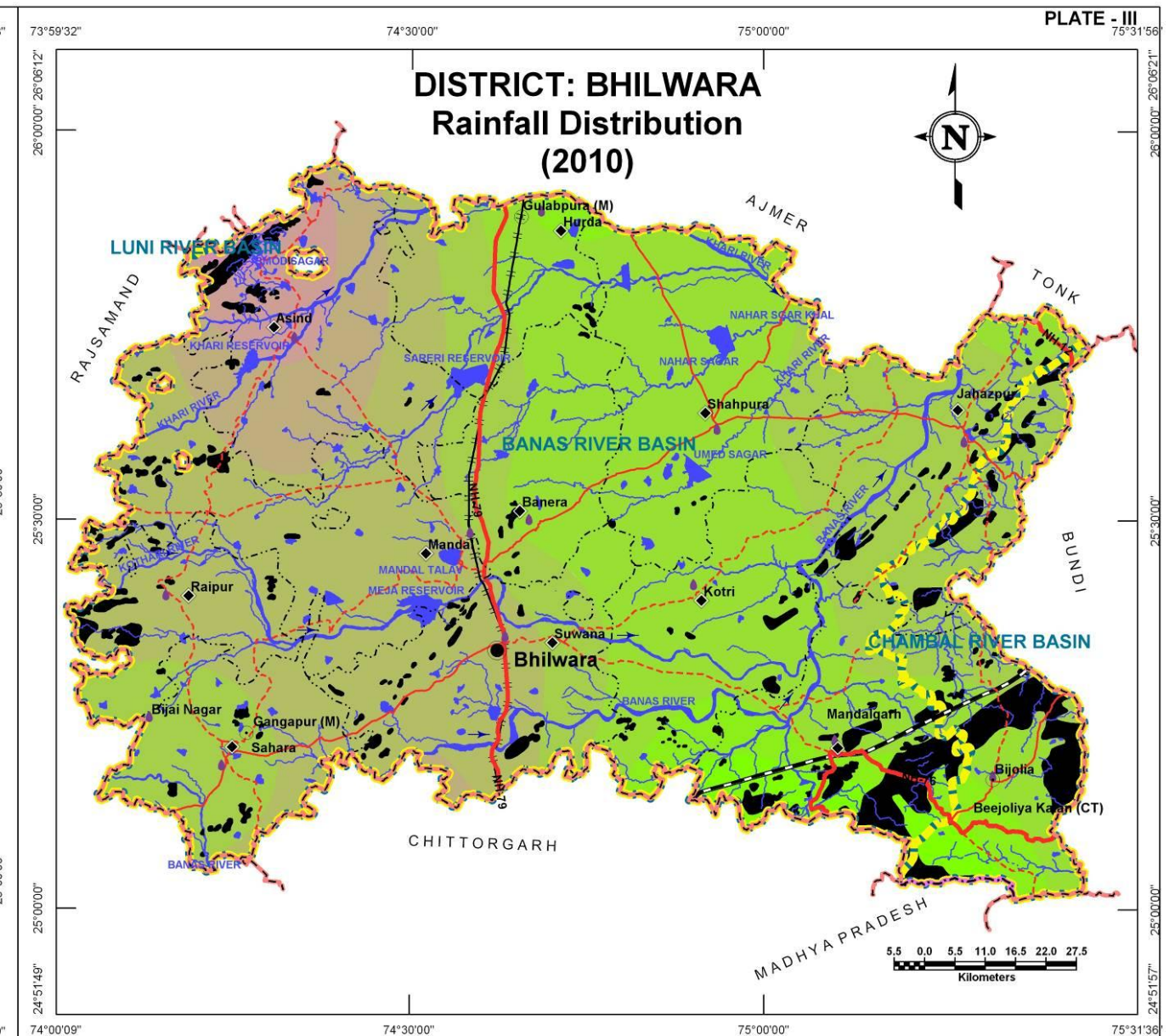
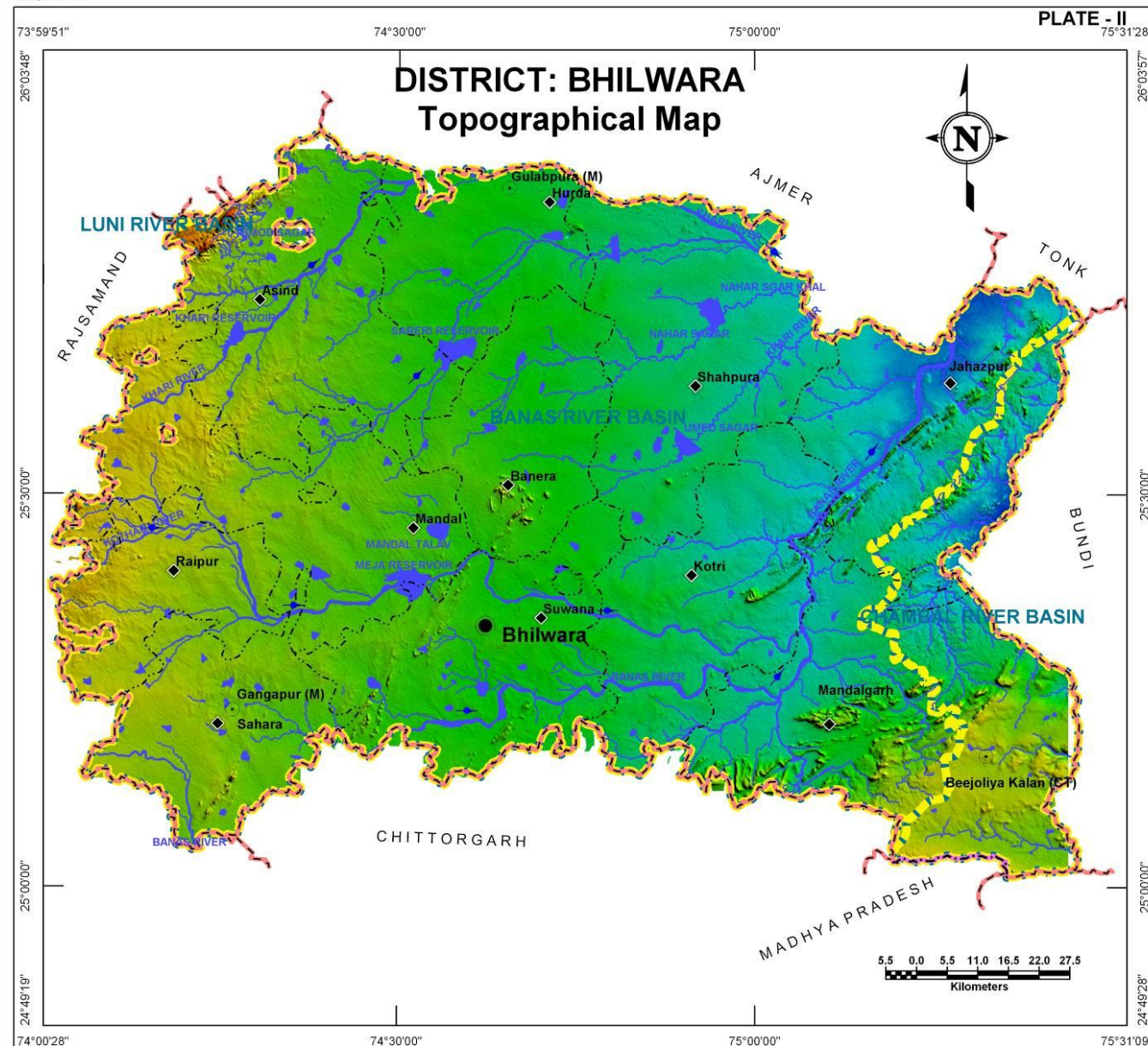
S. No.	Block Name	Minimum Elevation (m amsl)	Maximum Elevation (m amsl)
1	Asind	417.1	821.9
2	Banera	380.5	568.5
3	Hurda	328.7	453.3
4	Jahazpur	304.8	528.3
5	Kotri	322.1	492.1
6	Mandal	421.5	805.1
7	Mandalgarh	324.4	603.9
8	Raipur	465.9	610.3
9	Sahara	442.1	636.5
10	Shahpura	338.7	489.0
11	Suwana	380.7	550.0

RAINFALL

The rainfall received in the district is fairly good. The general distribution of rainfall across can be visualized from isohyets presented in the Plate – III where district is having uneven type of rainfall pattern. The rainfall isohyets follow north to south in the center part, generally receiving 800 to 900 mm while in the east while the west received 600 to 700 mm. Rainfall gradually decreases from southeast towards northwest. The annual average rainfall was 743.1 mm based on the data of available blocks. Mandalgarh block received highest annual rainfall (1281.8 mm) whereas minimum was in Asind block (347.0 mm). Highest average annual rainfall was recorded in Mandalgarh block which was about 901.6 mm.

Table: Block wise annual rainfall statistics (derived from year 2010 meteorological station data)

Block Name	Minimum Annual Rainfall (mm)	Maximum Annual Rainfall (mm)	Average Annual Rainfall (mm)
Asind	347.0	746.2	542.3
Banera	669.5	850.4	800.3
Hurda	575.7	916.6	770.5
Jahazpur	735.8	827.3	770.8
Kotri	754.6	995.9	851.9
Mandal	488.8	732.9	636.3
Mandalgarh	768.3	1,281.8	901.6
Raipur	631.1	735.5	676.8
Sahara	621.3	777.3	710.6
Shahpura	782.1	914.1	834.7
Suwana	582.3	848.5	677.8



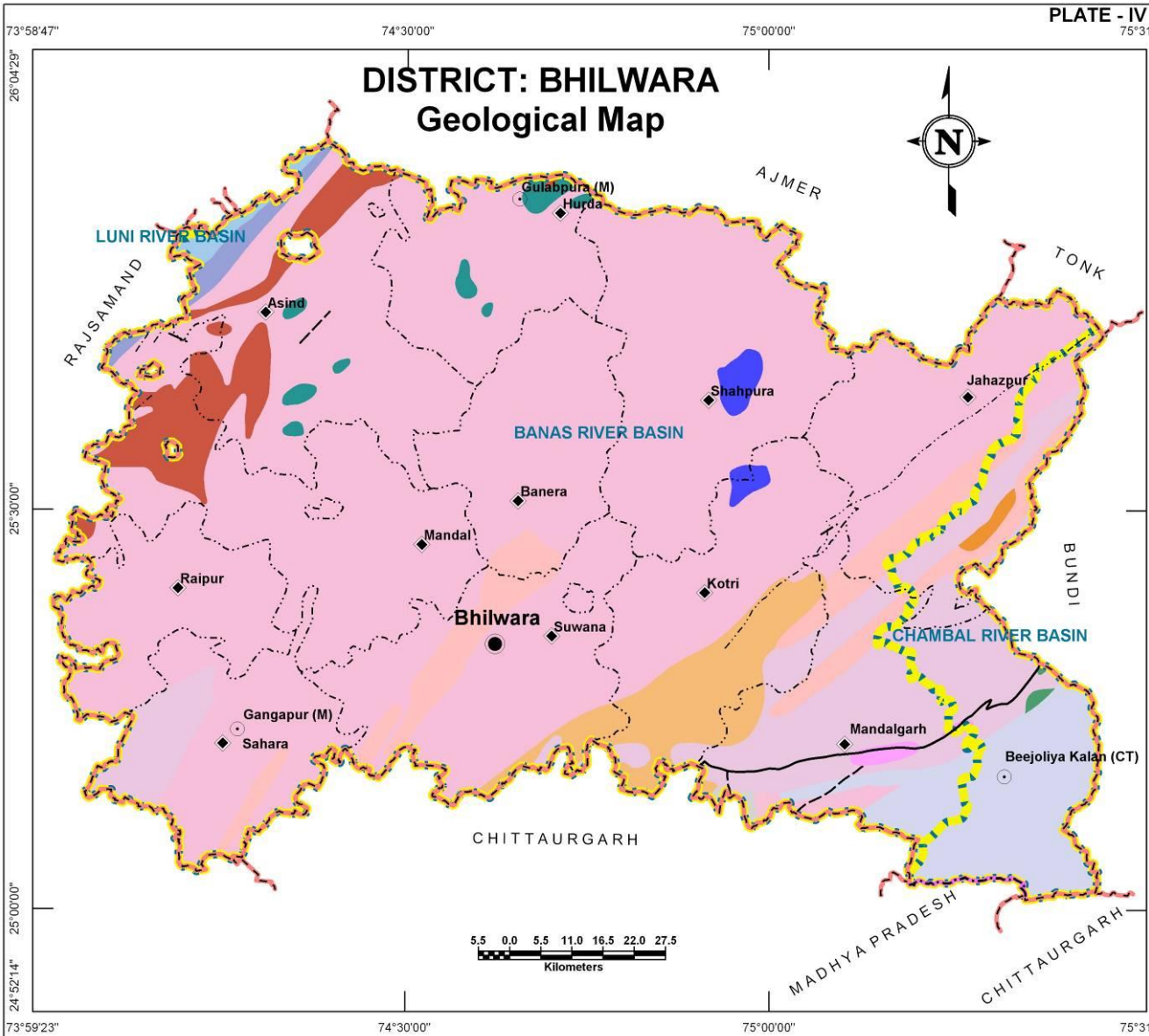
Geologically, Bhilwara Super Group occupies major part of the district. The Vindhyan Super Group represented by sandstone, shale and limestone is exposed in south-eastern part of the district along the great boundary fault. The Gogunda Group is exposed in extreme north-western part of the district and the Kumbhalgarh Group occupies small area in south-western part of the district, both belonging to Delhi Super Group. The Aravalli Super Group exposed in western part of the district is represented by Dovda Group.

Super Group	Group	Formation
	Recent	Alluvium, blown sand
Vindhyan	Bhander	Shale, sandstone and limestone
Delhi	Kumbhalgarh	Calc-Schist, gneisses, marble, garnet-boitite schist, migmatites
	Gogunda	Quartzite, biotite schist, Calc-schist, hornblende schist
Aravalli	Dovda	Quartzite, dolomite marble, dolomite, amphibolites, calc-silicates rocks, biotite schist, quartz-biotite gneiss and migmatites
	Igneous Intrusive	Dolerite dykes and sills
Bhilwara	Ranthambor	quartzite, shales and slates
	Igneous Intrusive	Berach granite and gneisses Jahajpur granite
	Pur-Banera	Calc-schist, gneisses, banded magnetite-quartzite, dolomitic marble, quartzite, garnet-mica schist and amphibolites
	Jahajpur	Dolomite, Phyllite, Quartzite
	Hindoli	Shales, slates, phyllites, metagrawacke, limestone, dolomite, quartzite, mica schist, meta basics, volcanics.
	Mangalwar Complex	Migmatites, gneisses, mica schist, garnetiferous mica schist, para-amphibolites silicified mica schist.
	Sandmata Complex	Para-gneisses, migmatites, pyroxene-granulites, amphibolites, biotite-ultramafic rocks.
	Igneous Intrusive	Acidic and mafic bodies

GEOMORPHOLOGY

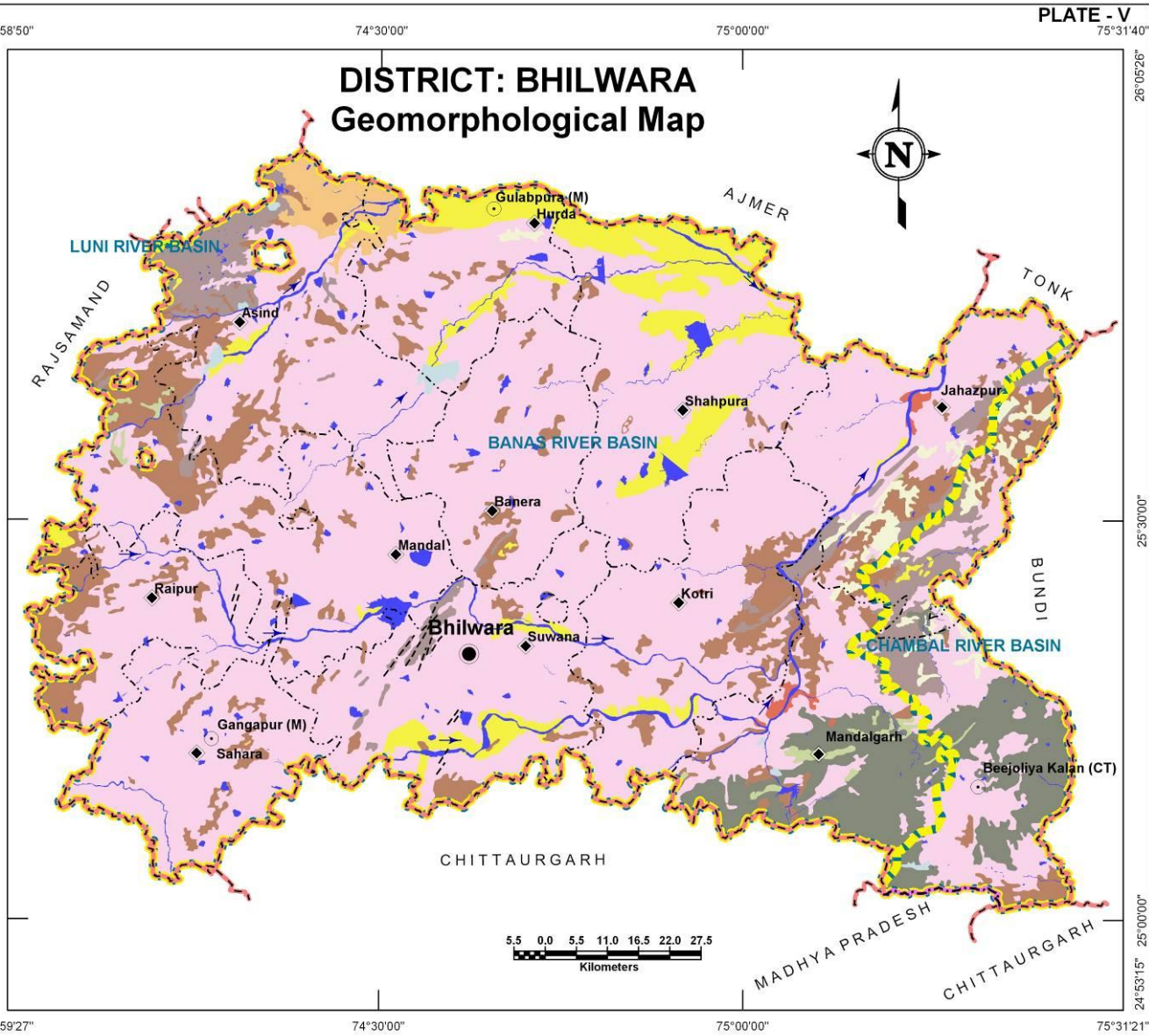
Table: Geomorphologic units, their description and distribution

Origin	Landform Unit	Description
Aeolian	Sandy Plain	Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt and clay.
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Intermontane Valley	Depression between mountains, generally broad & linear, filled with colluvial deposits.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
Fluvial	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
	Water logged/ Wetland	Area submerged in water or area having very shallow water table. So that it submerges in water during rainy season.
Structural	Plateau	Formed over varying lithology with extensive, flat, landscapes, bordered by escarpment on all sides. Essentially formed horizontally layered rocky marked by extensive flat top and steep slopes. It may be criss crossed by lineament.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



LEGEND			
Admin Boundary:		Geology:	
District Headquarter		Kumbhalgarh Group	
Block Headquarter		Gogunda Group	
Town		Dovda Group	
State Boundary		Ranthambhor Group	
District Boundary		Jahazpur Group	
Block Boundary		Pur-Banera Group	
Basin Boundary		Rajpura-Dariba Group	
Water Bodies		Hindoli Group	
Structural Features:		Mangalwar Complex	
Fault		Sand Mata Complex	
Great Boundary Fault		Bhander Group	
Shear		Rewa Group	
Thrust		Kaimur Group	

Source: District Resource Map of Rajasthan - GSI



LEGEND			
Admin Boundary:		Hills:	
District Headquarter		Structural/Denudational/Linear Ridge	
Block Headquarter			
Town			
State Boundary			
District Boundary			
Block Boundary			
Basin Boundary			
Water Bodies:			
River/Ponds/Reservoirs			
Structural Features:			
Fault/Fractures/Lineament			

Source: Ground Water Atlas of Rajasthan - SRSAC & GWD, Rajasthan

AQUIFERS

In this district, the aquifers are formed only within the weathered, fractured and jointed hardrock areas. Alluvial aquifers are not present in this district. Spatial distribution wise Schist is the most predominant aquifer type with 47.5% of the district area, Gneisses are next in the sequence with about 43% spatial coverage. Vindhyan sandstones occupy just less than 4% of the district area. Schistose aquifers are primarily formed in eastern part of the district, whereas the western part is primarily gneissic in nature and sandstones occupy the southeastern most part of the district.

DISTRICT – BHILWARA

Table: aquifer potential zones their area and their description

Aquifer in Potential Zone	Area (sq km)	% age of district	Description of the unit/Occurrence
Sandstone	403.0	3.9	Fine to medium grained, red colour and compact and at places.
Schist	4,960.6	47.5	Medium to fine grained compact rock. The litho units are soft, friable and have closely spaced cleavage.
Gneiss	4,463.4	42.7	Comprises of porphyritic and non-porphyritic gneissic complex.
Hills	618.1	5.9	-
Total	10,445.1	100.0	

STAGE OF GROUND WATER DEVELOPMENT

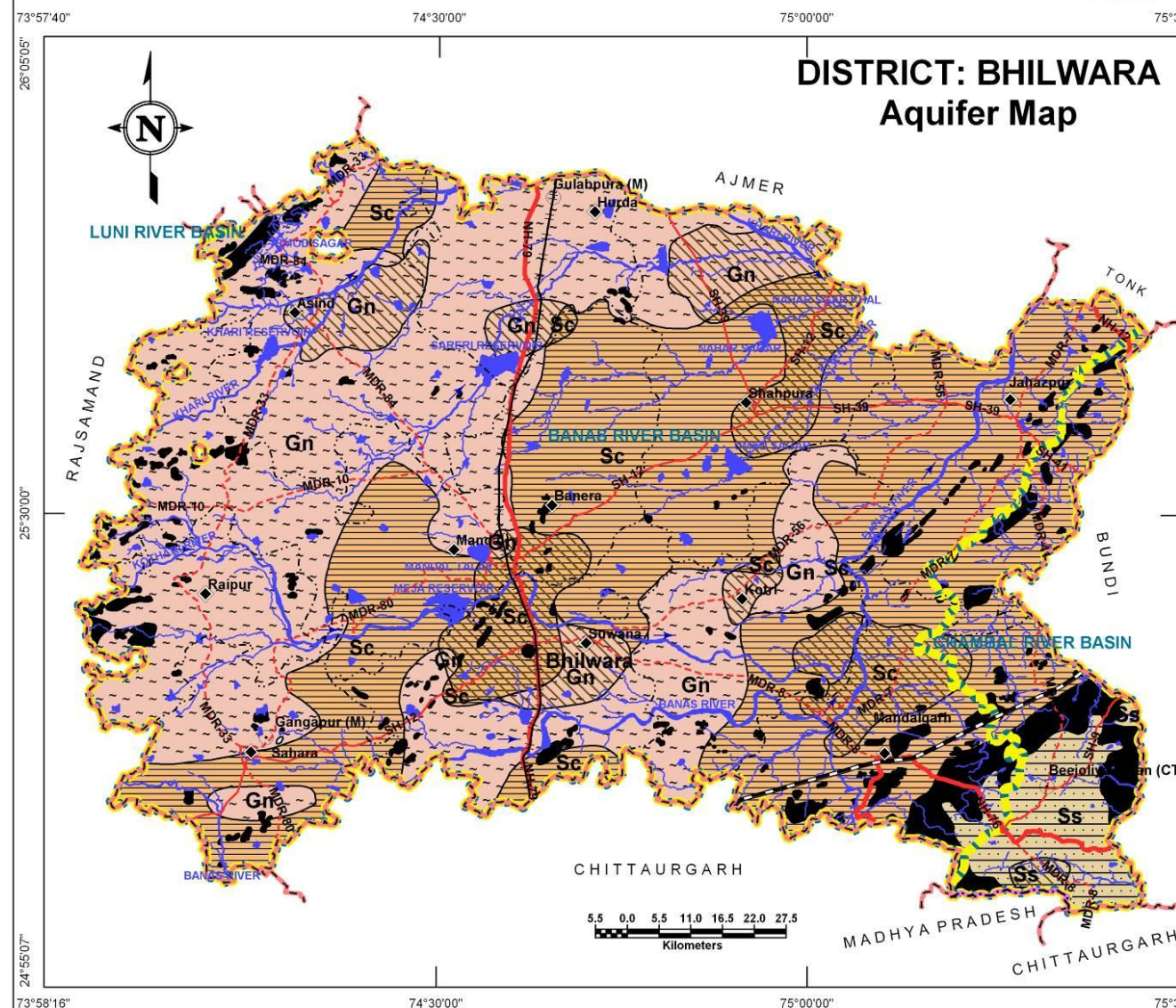
All the 11 blocks of the district are 'Over Exploited' since the aquifer development is complete and severe stress indicated on ground water resources since its exploitation has exceeded the recharge. As a result of this situation the ground water should not be extracted anymore and alternative means of fulfilling the needs be resorted too.

Categorization on the basis of stage of development of ground water	Block Name
Over Exploited	Raipur, Mandal, Kotri, Banera, Asind, Shahpura, Jahazpur, Mandalgarh, Suwana, Sahara, Hurda

Basis for categorization: Ground water development >100% - Over-Exploited.



PLATE -VI



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Roads:
 - National Highway
 - State Highway
 - Major District Road
- Railway:
 - Broad Gauge
 - Metre Gauge

Water Bodies:

- River / Streams
- Ponds / Reservoirs

Hills

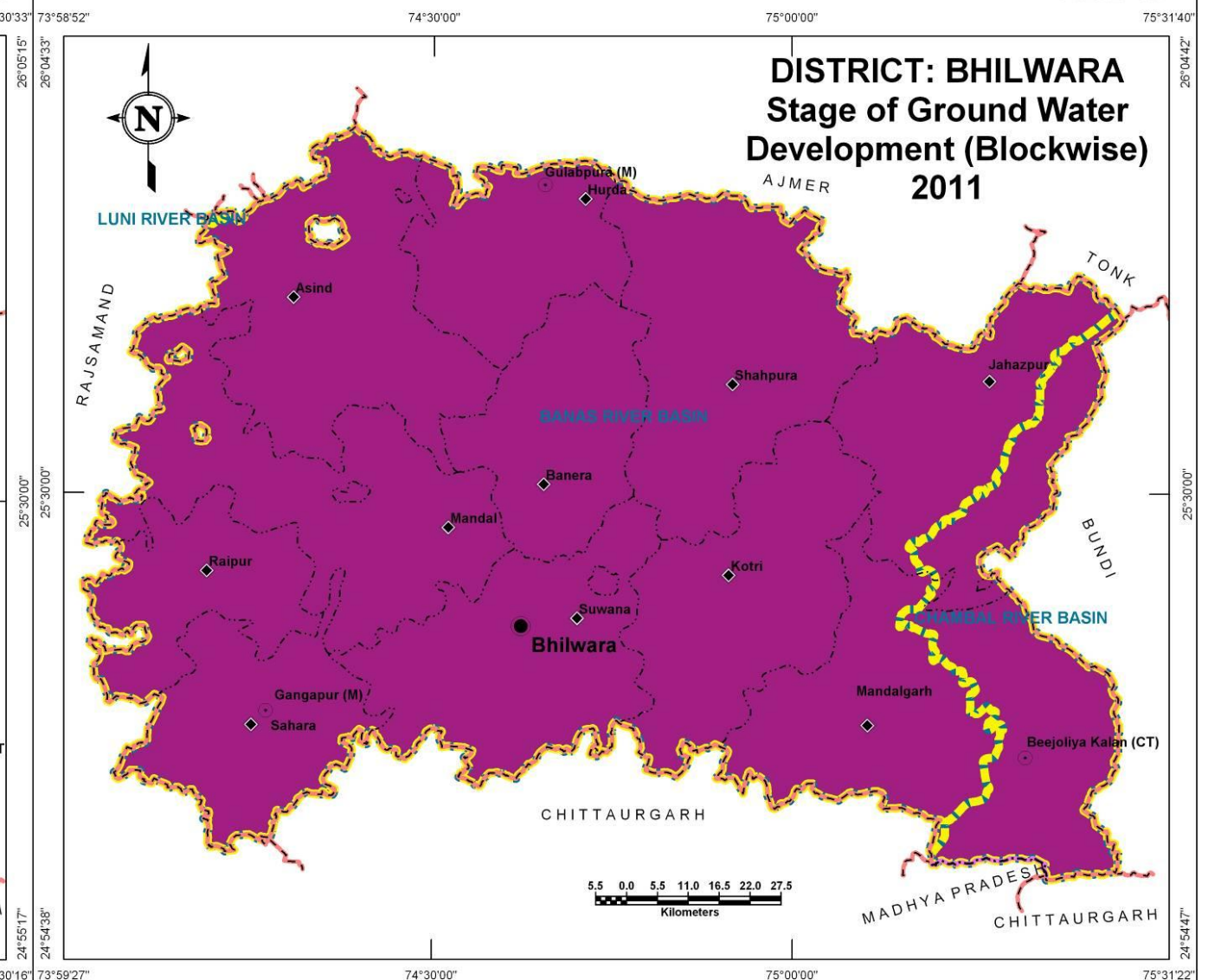
Command Area

Aquifer:

- Sandstone
- Schist
- Gneiss

Source: Ground Water Potential Zone Map - GWD, Rajasthan

PLATE - VII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary

Stage of Ground Water Development:

- Over-Exploited

Source: Ground Water Department, Rajasthan

LOCATION OF EXPLORATORY AND GROUNDWATER MONITORING WELLS

DISTRICT – BHILWARA

Bhilwara district has a well distributed network of exploratory wells (85) and groundwater monitoring stations (355) in the district owned by RGWD (72 and 310 respectively) and CGWB (13 and 45 respectively). The exploratory wells have formed the basis for delineation of subsurface aquifer distribution scenario in three dimensions. Benchmarking and optimization studies suggest that ground water level monitoring network is sufficiently distributed for appropriate monitoring and for water quality, two wells in Mandalgarh block are recommended to be added to existing network for optimum monitoring of the aquifers.

Table: Block wise count of wells (existing and recommended)

Block Name	Exploratory Wells			Ground Water Monitoring Stations			Recommended additional wells for optimization of monitoring network	
	CGWB	RGWD	Total	CGWB	RGWD	Total	Water Level	Water Quality
Asind	-	6	6	5	30	35	-	-
Banera	-	5	5	5	29	34	-	-
Hurda	-	6	6	2	26	28	-	-
Jahazpur	1	7	8	4	28	32	-	-
Kotri	-	7	7	5	28	33	-	-
Mandal	-	9	9	1	37	38	-	-
Mandalgarh	-	9	9	5	26	31	-	2
Raipur	-	6	6	4	20	24	-	-
Sahara	-	5	5	2	26	28	-	-
Shahpura	11	5	16	5	25	30	-	-
Suwana	1	7	8	7	35	42	-	-
Total	13	72	85	45	310	355	-	2

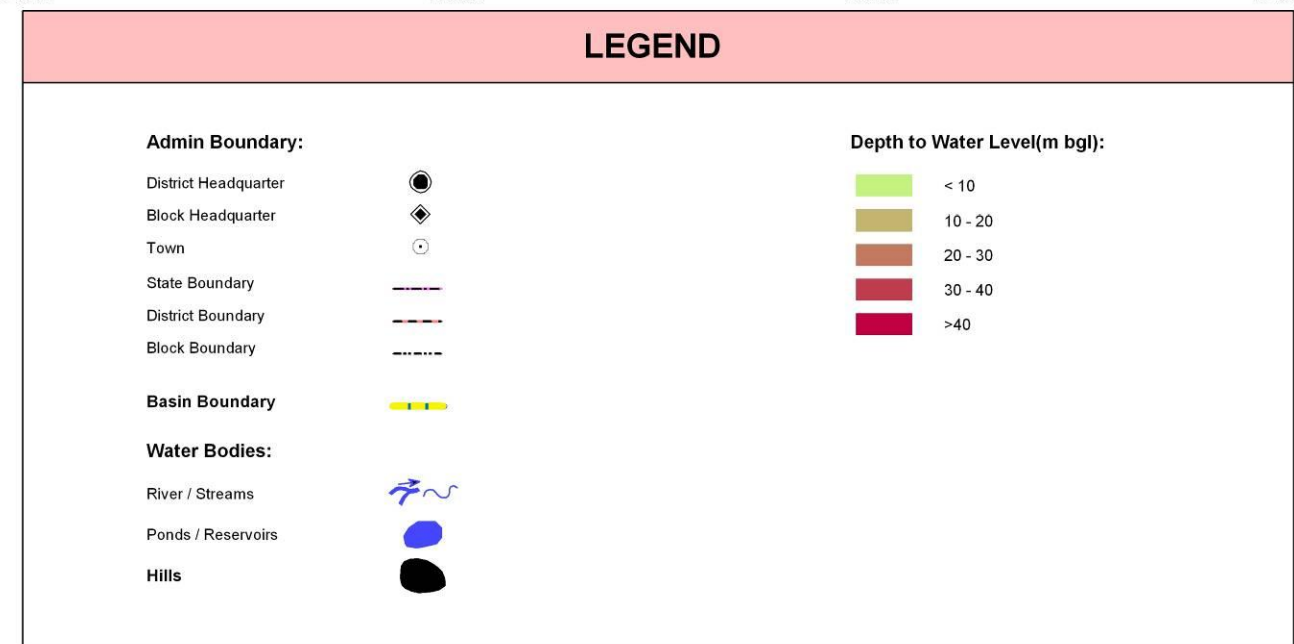
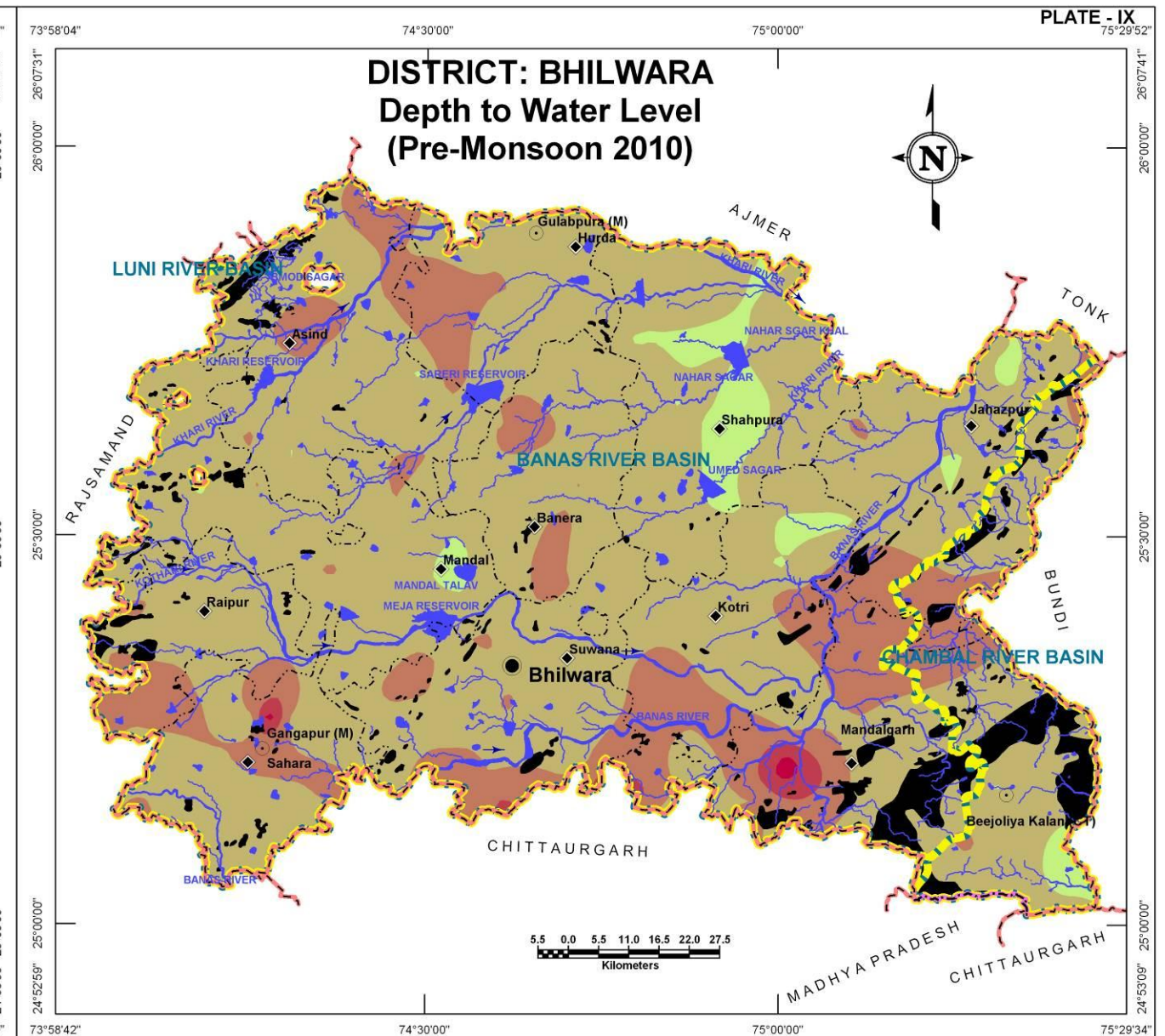
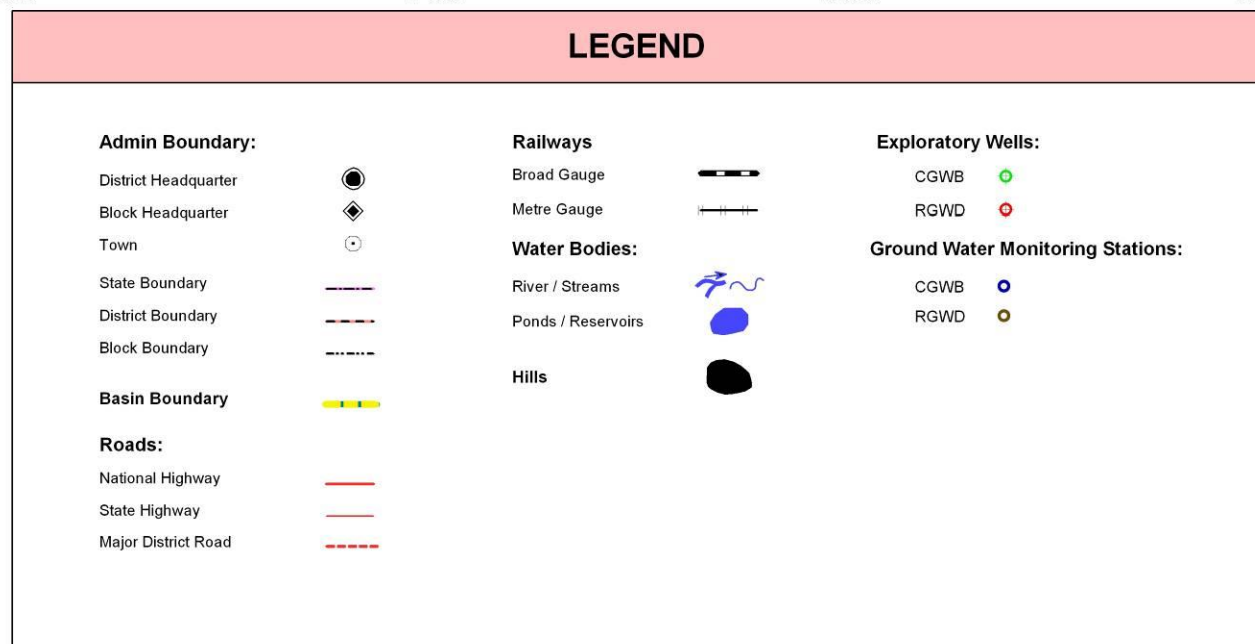
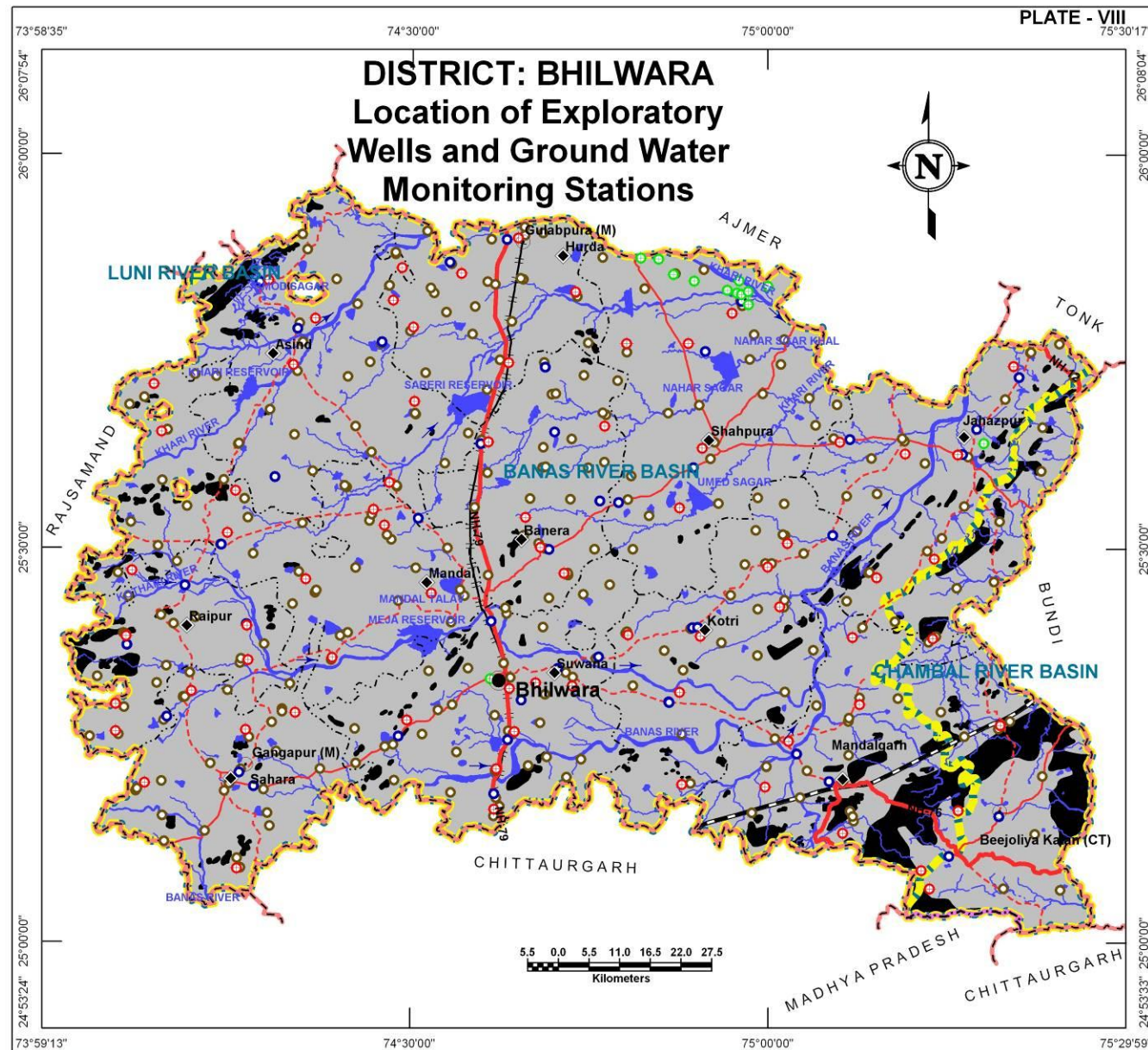
DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

In spite of being a predominantly hard rock area, the district shows moderate variation in depth to ground water levels from less than 10m bgl to more than 40m bgl. Shallow water levels less than 10m bgl are seen in isolated patches north and south eastern parts of Jahazpur, Kotri, Mandalgarh, Shahpura and central part of Mandal blocks. Deeper water levels of more than 40m bgl are found in very small patches in Mandalgarh block and Gangapur town. Apart from these highs and lows, the depth to water level varies in between 10m bgl to 30m bgl in rest of the district area.

Table: Block wise area of depth to water level occurrence

Depth to water level (m bgl)	Block wise area coverage (sq km) *											Total Area (sq km)
	Asind	Banera	Hurda	Jahazpur	Kotri	Mandal	Mandalgarh	Raipur	Sahara	Shahpura	Suwana	
< 10	0.9	-	-	27.8	42.2	30.4	45.6	-	-	224.5	-	371.4
10-20	892.9	576.6	453.2	790.8	597.5	1,059.9	684.7	374.2	505.6	924.0	738.3	7,597.7
20-30	170.3	106.7	159.5	183.6	276.3	13.0	402.3	129.4	155.0	-	180.4	1,776.5
30-40	2.1	-	0.1	-	4.5	-	48.7	-	12.4	-	5.7	73.5
> 40	-	-	-	-	-	-	7.3	-	0.6	-	-	7.9
Total	1,066.2	683.3	612.8	1,002.2	920.5	1,103.3	1,188.6	503.6	673.6	1,148.5	924.4	9,827.0

* The area covered in the derived maps is less than the total district area since the hills have been excluded from interpolation/contouring.



WATER TABLE ELEVATION (PRE MONSOON – 2010)

DISTRICT – BHILWARA

General flow direction of groundwater is from west to east in most parts of the district and from southeast to north in the southeastern part of the district. Highest water table elevation (>560m amsl) is seen in the western part (Raipur Block) of the district which gradually lowers towards east where the minimum elevation (<260m amsl) is seen in the Jahazpur block. The general elevation in the district ranges between 320m to 520m amsl. Flow gradient is steep in the hilly part of southeastern Bhilwara but moderate in rest of the district.

Table: Block wise area covered in each water table elevation range

Water table elevation (m amsl) range	Block wise area coverage (sq km)											Total Area (sq km)
	Asind	Banera	Hurda	Jahazpur	Kotri	Mandal	Mandalgarh	Raipur	Sahara	Shahpura	Suwana	
< 260	-	-	-	-	-	0.1	-	-	-	-	-	0.1
260 - 280	-	-	-	-	-	0.9	-	-	-	-	-	0.9
280 - 300	-	-	-	-	-	3.0	-	-	-	-	-	3.0
300 - 320	-	-	-	281.5	-	4.7	6.8	-	0.5	-	-	293.5
320 - 340	-	-	-	670.1	234.8	4.9	230.6	-	3.0	25.8	-	1,169.2
340 - 360	-	-	-	50.6	381.3	8.1	239.8	-	4.0	548.2	-	1,232.0
360 - 380	-	60.1	193.2	-	209.8	16.2	170.4	-	1.5	574.3	42.4	1,267.9
380 - 400	-	325.2	157.7	-	94.6	26.5	86.3	0.1	1.6	0.2	470.0	1,162.2
400 - 440	432.0	298.0	261.9	-	-	290.4	101.1	8.8	10.7	-	392.7	1,795.6
440 - 480	401.3	-	-	-	-	315.4	187.1	123.5	518.7	-	19.3	1,565.3
480 - 520	214.0	-	-	-	-	168.4	166.5	269.8	133.6	-	-	952.3
520 - 560	18.9	-	-	-	-	247.8	-	99.9	-	-	-	366.6
> 560	-	-	-	-	-	16.9	-	1.5	-	-	-	18.4
Total	1,066.2	683.3	612.8	1,002.2	920.5	1,103.3	1,188.6	503.6	673.6	1,148.5	924.4	9,827.0

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

The ground water level fluctuation map (Plate – XI) reveals a fall of 4 m in some areas to rise in other areas reaching upto 18m. The –ve fluctuation areas (indicated by pink and red regions) are the areas where overexploitation is taking place. Such large ground water depletion areas are very localized and seen in the southern and southeastern part of the district. Rest of the district has shown a general rise between 2 to 12m. Maximum rise of more than 18m is noticed at northern part of Mandalgarh block.

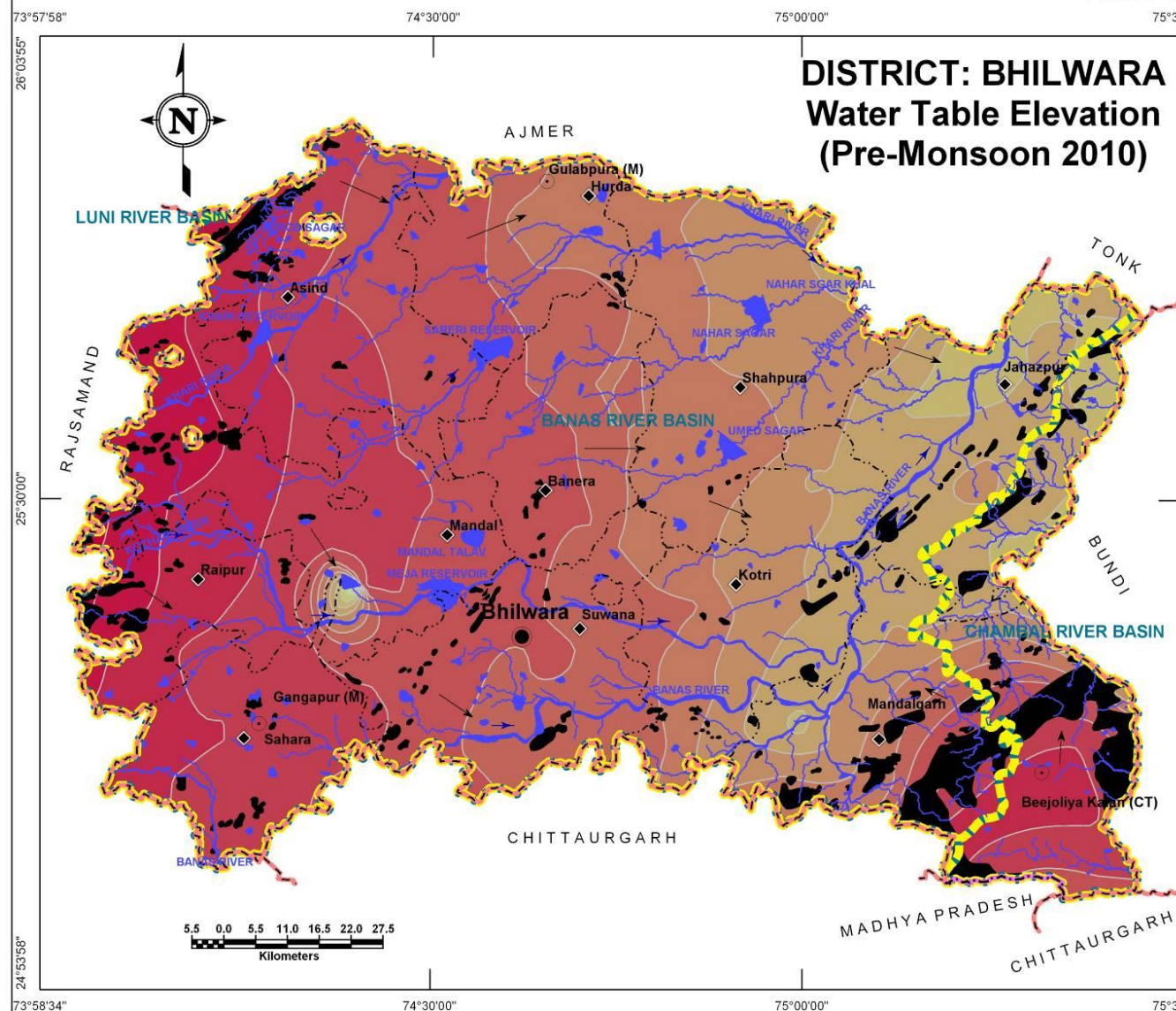
Table: Block wise area covered in each water fluctuation zone

Water level fluctuation range (m)	Block wise area coverage (sq km)											Total Area (sq km)
	Asind	Banera	Hurda	Jahazpur	Kotri	Mandal	Mandalgarh	Raipur	Sahara	Shahpura	Suwana	
< -4	-	-	-	-	-	-	3.7	-	-	-	-	3.7
-4 – -2	-	-	-	-	-	-	12.5	-	-	1.2	4.3	18.0
-2 – 0	-	2.6	-	-	-	-	26.8	-	-	11.5	23.8	64.7
0 – 2	48.7	5.6	-	3.8	-	14.3	117.5	31.3	6.9	68.3	52.6	349.0
2 – 4	203.3	19.0	1.3	41.2	46.7	190.4	98.8	33.5	31.1	487.5	148.9	1,301.7
4 – 6	161.1	157.8	29.4	230.1	166.6	229.2	53.5	137.5	103.7	301.0	271.4	1,841.3
6 – 8	220.7	160.9	74.5	282.9	276.7	246.7	93.9	143.3	183.4	162.8	262.5	2,108.3
8 – 10	205.0	118.6	175.3	154.4	242.9	339.6	183.6	61.5	219.1	74.3	107.3	1,881.6
10 – 12	129.5	109.7	200.1	137.7	112.7	67.3	226.9	46.3	91.1	30.2	32.7	1,184.2
12 – 14	65.4	74.7	104.8	106.9	51.3	12.4	171.2	25.8	30.9	11.7	13.9	669.0
14 – 16	27.3	25.9	20.3	31.8	22.5	3.0	70.6	18.6	7.4	-	7.0	234.4
16 – 18	4.6	7.4	7.1	10.5	1.1	0.4	69.3	5.6	-	-	-	106.0
> 18	0.6	1.1	-	2.9	-	-	60.3	0.2	-	-	-	65.1
Total	1,066.2	683.3	612.8	1,002.2	920.5	1,103.3	1,188.6	503.6	673.6	1,148.5	924.4	9,827.0



PLATE - X

DISTRICT: BHILWARA
Water Table Elevation
(Pre-Monsoon 2010)



LEGEND

Admin Boundary:

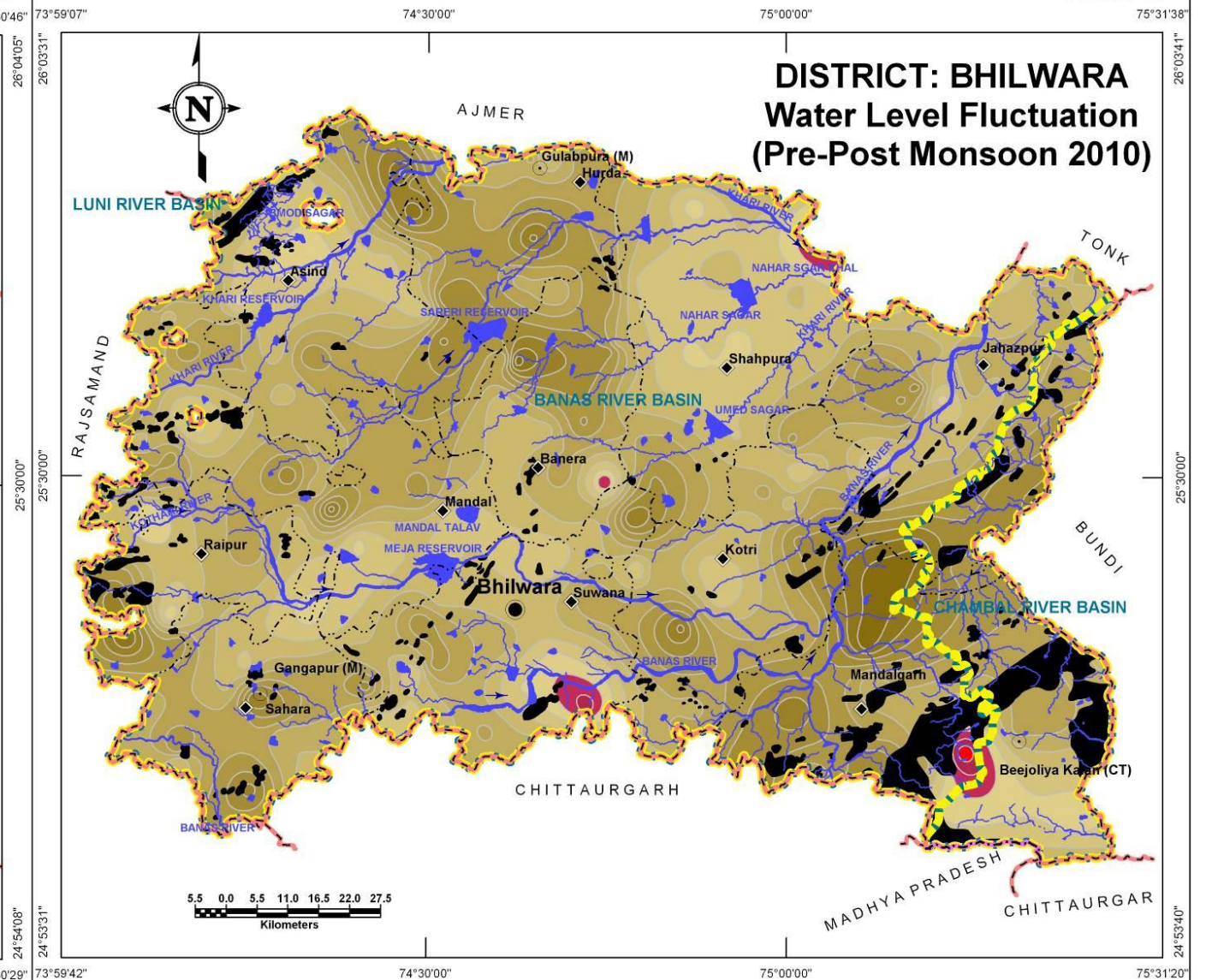
- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
- River / Streams
- Ponds / Reservoirs
- Hills
- Water Table Elevation

Water Table Elevation(m amsl):

- <260
- 260 - 280
- 280 - 300
- 300 - 320
- 320 - 340
- 340 - 360
- 360 - 380
- 380 - 400
- 400 - 440
- 440 - 480
- 480 - 520
- 520 - 560
- >560

PLATE - XI

DISTRICT: BHILWARA
Water Level Fluctuation
(Pre-Post Monsoon 2010)



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Water Bodies:
- River / Streams
- Ponds / Reservoirs
- Hills

Water Level Fluctuation(m):

- <-4
- 4 to -2
- 2 to 0
- 0 to 2
- 2 to 4
- 4 to 6
- 6 to 8
- 8 to 10
- 10 to 12
- 12 to 14
- 14 to 16
- 16 to 18
- > 18

GROUND WATER ELECTRICAL CONDUCTIVITY DISTRIBUTION

DISTRICT – BHILWARA

The Electrical conductivity (at 25°C) distribution map is presented in Plate – XII. The areas with low electrical conductivity of ground water (>2000 $\mu\text{S}/\text{cm}$) are shown in yellow color and occupies approximately 60% of the district area which is suitable for domestic purpose. The areas with moderately high EC Values (2000-4000 $\mu\text{S}/\text{cm}$) are shown in green color and occupy approximately 30% of the district. Remaining small portion of the district area (approximately 5%) falls under high EC values (<4000 $\mu\text{S}/\text{cm}$) area which is shown in red color, largely in the northern, central and southern part of the district where the ground water is unsuitable for domestic purpose.

Table: Block wise area of Electrical conductivity distribution

Electrical Conductivity Ranges (μS/cm at 25°C) (Ave. of years 2005-09)	Block wise area coverage (sq km)																						Total Area (sq km)
	Asind		Banera		Hurda		Jahazpur		Kotri		Mandal		Mandalgarh		Raipur		Sahara		Shahpura		Suwana		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 2000	535.9	50.2	486.6	71.2	79.6	13.0	693.8	69.2	763.9	82.9	821.4	74.4	1,188.6	100.0	255.7	50.8	386.9	57.4	208.7	18.2	498.4	54.0	5,919.5
2000-4000	491.1	46.1	196.7	28.8	482.9	78.8	282.8	28.2	155.2	16.9	266.7	24.2	-	-	226.2	44.9	264.7	39.3	721.1	62.8	341.5	37.0	3,428.9
>4000	39.2	3.7	-	-	50.3	8.2	25.6	2.6	1.4	0.2	15.2	1.4	-	-	21.7	4.3	22.0	3.3	218.7	19.0	84.5	9.0	478.6
Total	1,066.2	100.0	683.3	100.0	612.8	100.0	1,002.2	100.0	920.5	100.0	1,103.3	100.0	1,188.6	100.0	503.6	100.0	673.6	100.0	1,148.5	100.0	924.4	100.0	9,827.0

GROUND WATER CHLORIDE DISTRIBUTION

High chloride concentration in ground water also renders it unsuitable for domestic and other purposes. The green colored regions in Plate – XIII are such areas where chloride concentration is moderately high (250-1000 mg/l) which occupy about 55% of the district area. The areas with low chloride concentration (<250 mg/l) are shown in yellow color and occupy approximately 41% of the district area, largely in the eastern and southeastern part of the district. Remaining small part of the district approximately 3% falls under high chloride concentration (>1000 mg/l) area which is shown in red color, largely south of Bhilwara city and around Shahpura. The ground water in this region is unsuitable for domestic purpose. Overall, the southeastern part has low chloride concentration in ground water whereas the rest of the area has mixed distribution of moderate and low concentration in ground water.

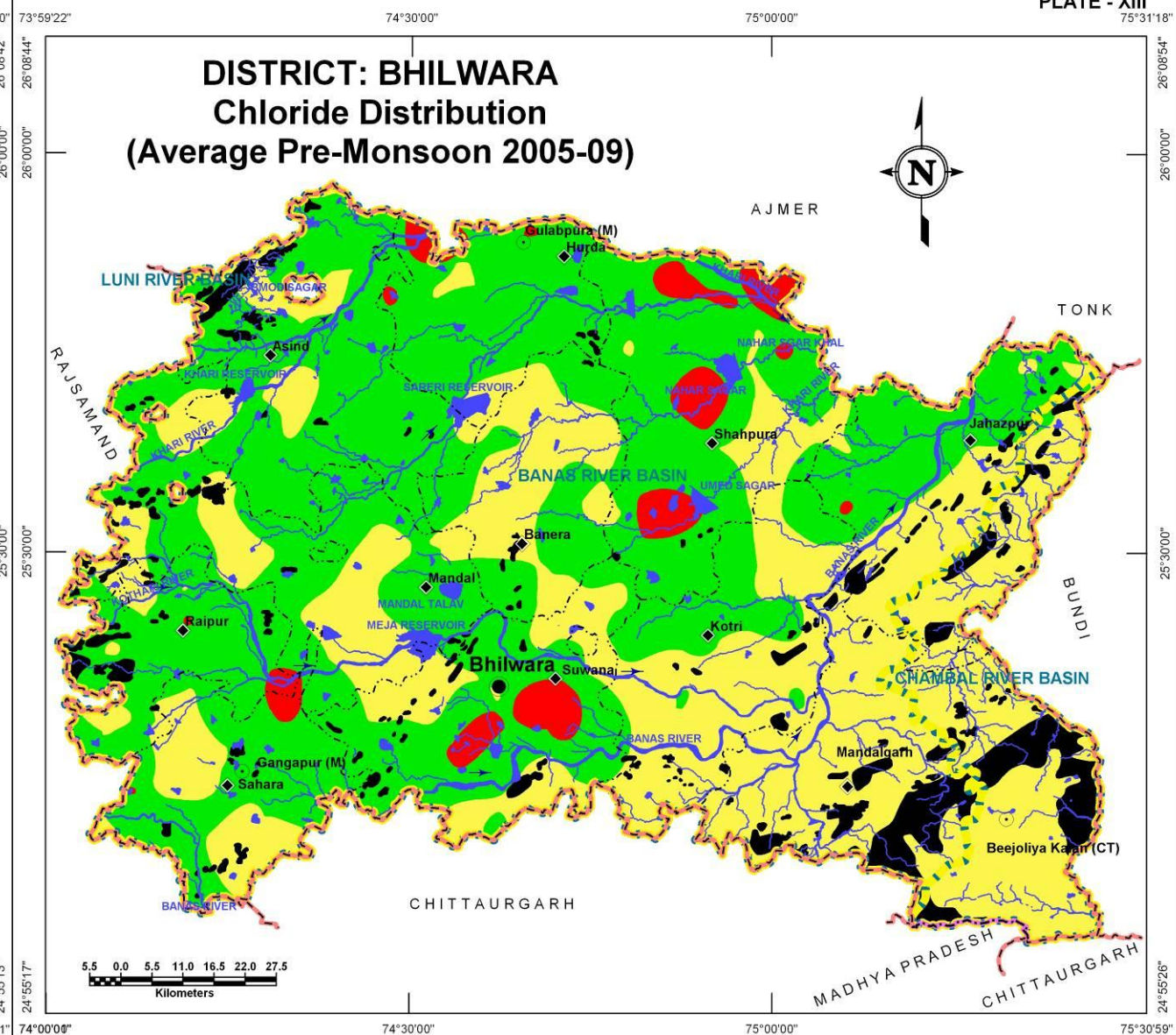
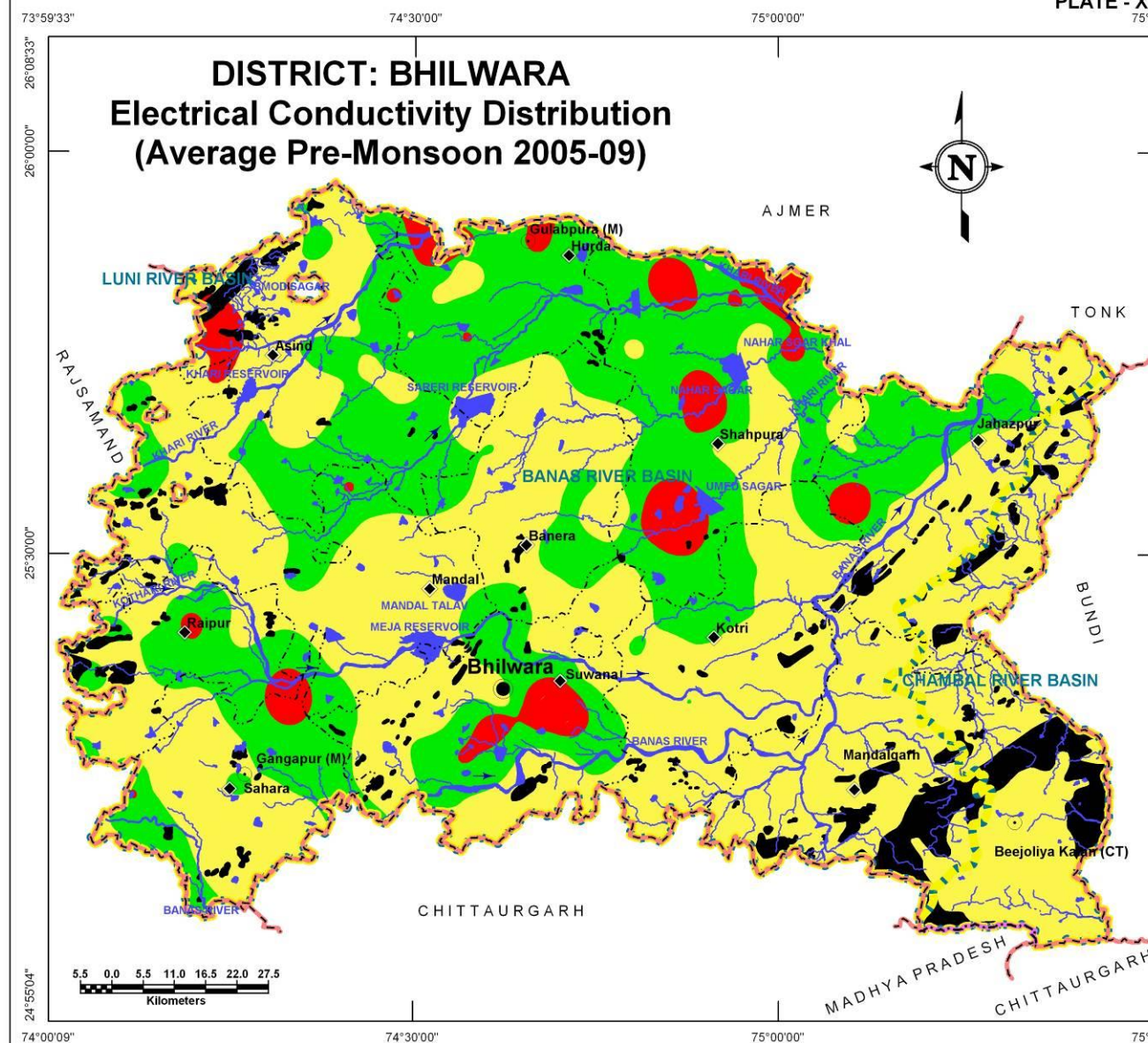
Table: Block wise area of Chloride distribution

Chloride Concentration Range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																						Total Area (sq km)
	Asind		Banera		Hurda		Jahazpur		Kotri		Mandal		Mandalgarh		Raipur		Sahara		Shahpura		Suwana		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 250	167.8	16.0	359.1	53.0	53.4	9.0	547.1	55.0	619.2	67.0	411.7	37.0	1,188.6	100.0	57.1	11.0	220.4	33.0	142.3	12.0	266.4	29.0	4,033.1
250-1000	898.4	84.0	324.2	47.0	526.7	86.0	452.4	45.0	301.3	33.0	689.7	63.0	-	-	432.4	86.0	435.2	64.0	825.9	72.0	563.8	61.0	5,450.0
> 1000	-	-	-	-	32.7	5.0	2.7	-	-	-	1.9	-	-	-	14.1	3.0	18.0	3.0	180.3	16.0	94.2	10.0	343.9
Total	1,066.2	100.0	683.3	100.0	612.8	100.0	1,002.2	100.0	920.5	100.0	1,103.3	100.0	1,188.6	100.0	503.6	100.0	673.6	100.0	1,148.5	100.0	924.4	100.0	9,827.0



PLATE - XII

PLATE - XIII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Electrical Conductivity ($\mu\text{S}/\text{cm}$ at 25°C):

- < 2000
- 2000-4000
- > 4000

LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Chloride Concentration (mg/l):

- < 250
- 250 - 1000
- > 1000

GROUND WATER FLUORIDE DISTRIBUTION

DISTRICT – BHILWARA

The Fluoride concentration map is presented in Plate XIV. The areas with moderately high concentration (1.5–3.0 mg/l) are shown in green color and occupy approximately 43% of the district area. The areas with low concentration (i.e., >1.5 mg/l) are shown in yellow color which occupy 47% of the district area which is suitable for domestic purpose. Remaining part of the district approximately 10% has high Fluoride concentration (>3.0 mg/l) which is shown in red color, largely northern and northwestern part of the district. The ground water in this region is not suitable for domestic purpose.

Table: Block wise area of Fluoride distribution

Fluoride concentration range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																						Total Area (sq km)
	Asind		Banera		Hurda		Jahazpur		Kotri		Mandal		Mandalgarh		Raipur		Sahara		Shahpura		Suwana		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
<1.5	268.8	25.2	217.5	31.8	125.2	20.4	572.6	57.1	683.0	74.2	566.7	51.4	1,109.6	93.3	208.9	41.5	167.7	24.9	62.8	5.5	630.0	68.0	4,612.
1.5-3.0	699.6	65.6	417.6	61.1	408.8	66.7	298.4	29.8	201.7	21.9	517.2	46.8	76.8	6.5	292.1	58.0	503.1	74.7	561.6	48.9	277.7	30.0	4,254.
>3.0	97.8	9.2	48.2	7.1	78.8	12.9	131.2	13.1	35.8	3.9	19.4	1.8	2.2	0.2	2.6	0.5	2.8	0.4	524.1	45.6	16.7	2.0	959.
Total	1,066.2	100.0	683.3	100.0	612.8	100.0	1,002.2	100.0	920.5	100.0	1,103.3	100.0	1,188.6	100.0	503.6	100.0	673.6	100.0	1,148.5	100.0	924.4	100.0	9,827.

GROUND WATER NITRATE DISTRIBUTION

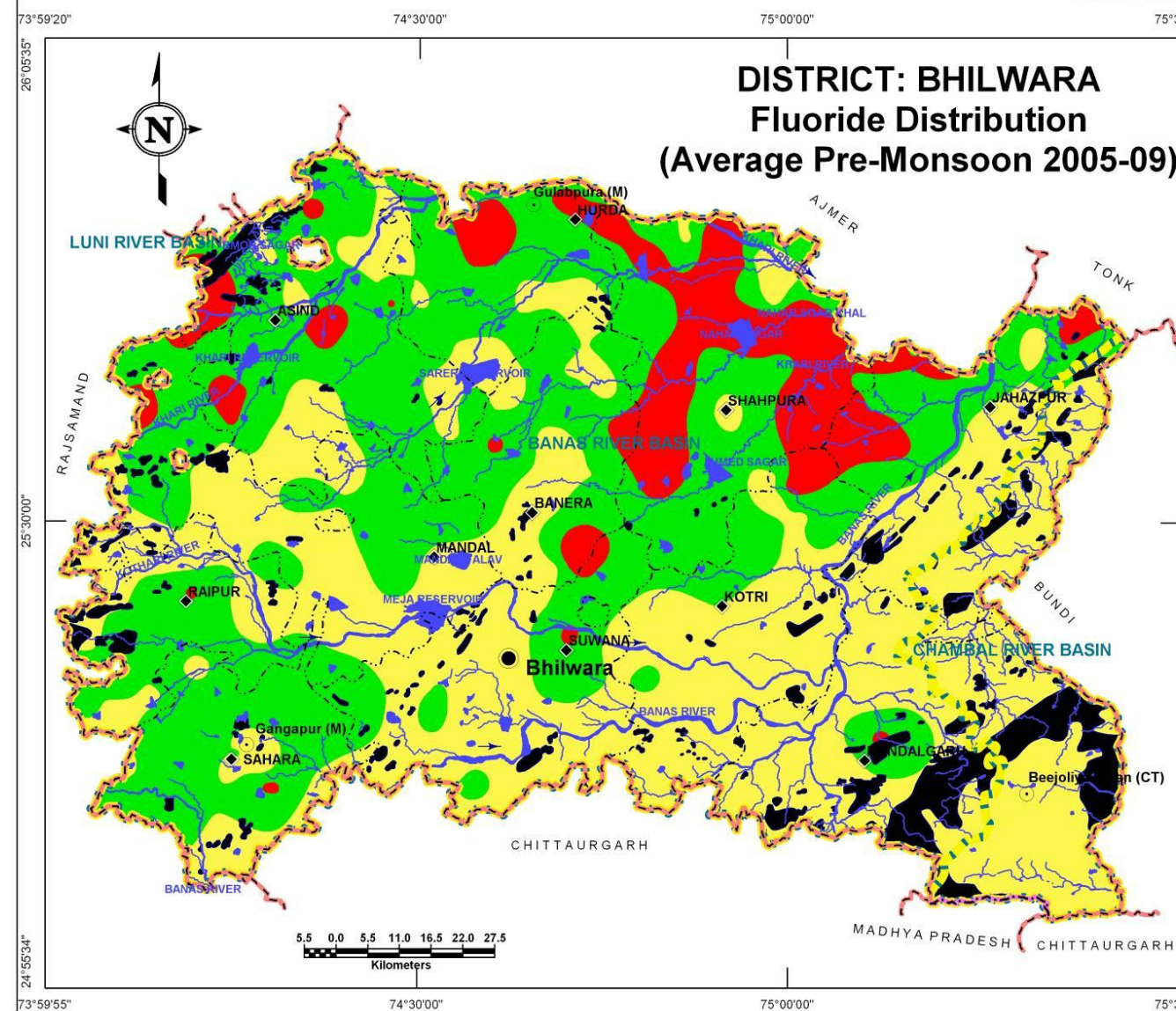
High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate – XV shows distribution of nitrate in groundwater. High nitrate concentration (>100 mg/l) is shown in red color and occupies approximately 32% of the district area which is not suitable for agriculture purpose. The distribution of such high nitrate areas does not appear to follow any systematic pattern as the large and small isolated patches are distributed all over the district. The areas with moderately high nitrate concentration (50-100 mg/l) are shown in green color and occupy approximately 33% of the district area, largely northern and northwestern part of the district. Remaining part of the district area approximately 35% is covered with low nitrate concentration (<50 mg/l) is shown in yellow colored patches. The ground water in this region is suitable for agriculture purpose.

Table: Block wise area of Nitrate distribution

Nitrate concentration range (mg/l) (Ave. of years 2005-09)	Block wise area coverage (sq km)																						Total Area (sq km)
	Asind		Banera		Hurda		Jahazpur		Kotri		Mandal		Mandalgarh		Raipur		Sahara		Shahpura		Suwana		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
<50	427.7	40.1	217.3	31.8	240.0	39.2	469.6	46.9	253.9	27.6	361.1	32.7	643.5	54.1	65.4	13.0	239.2	35.5	207.3	18.1	301.0	33.0	3,426.6
50-100	405.3	38.0	256.2	37.5	287.6	46.9	299.8	29.9	325.8	35.4	315.2	28.6	248.3	20.9	74.3	14.7	155.1	23.0	565.0	49.1	343.2	37.0	3,275.8
>100	233.2	21.9	209.8	30.7	85.2	13.9	232.8	23.2	340.8	37.0	427.0	38.7	296.8	25.0	363.9	72.3	279.3	41.5	376.2	32.8	280.2	30.0	3,125.2
Total	1,066.2	100.0	683.3	100.0	612.8	100.0	1,002.2	100.0	920.5	100.0	1,103.3	100.0	1,188.6	100.0	503.6	100.0	673.6	100.0	1,148.5	100.0	924.4	100.0	9,827.0



PLATE - XIV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

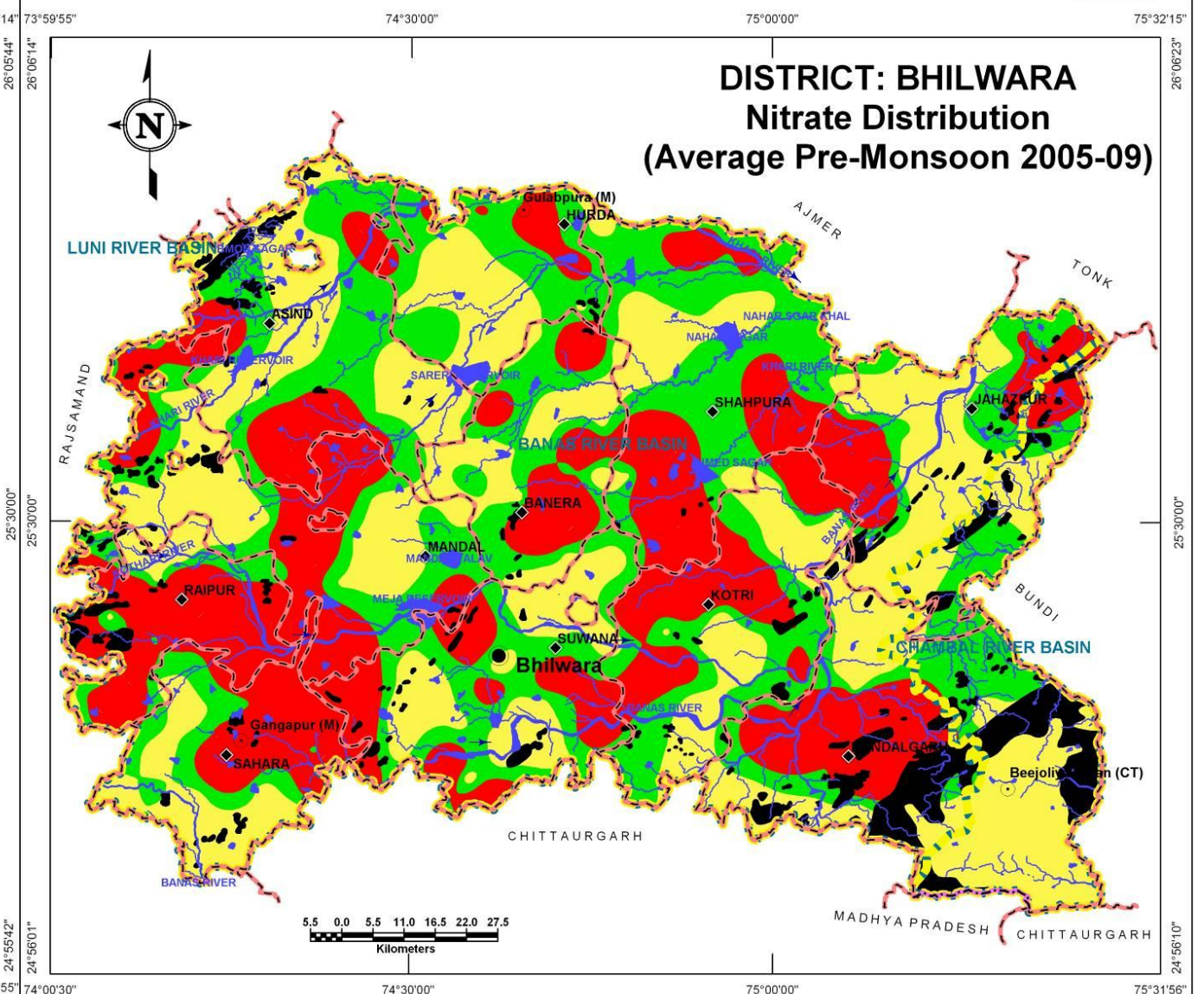
Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Fluoride Concentration (mg/l):

- < 1.5
- 1.5-3.0
- > 3.0

PLATE - XV



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Nitrate Concentration (mg/l):

- < 50
- 50-100
- > 100

DEPTH TO BEDROCK

DISTRICT – BHILWARA

Plate – XVI depicts the bedrock depth from ground level in Bhilwara district. The beginning of massive bedrock has been considered for defining top of bedrock surface. It varies from less than 20 below ground level to more than 80m bgl. The major rock types constituting the bedrock are sandstone, schist and gneisses. These rocks are overlain by alluvial deposits of sand, clay, silt and admixture of these in different proportions and thicknesses. The map of depth to bedrock in meters below ground level reveals that the bedrock surface is quite undulating and varies from very shallow near hardrock exposures to more than 80m below ground level. Bedrocks at very shallow depth (less than 20m bgl to 40m bgl) found in all blocks of the districts in patches and mostly in the north, southwest and eastern parts. Eastern part of the district in Raipur and Sahara blocks, the bedrock reaches a depth of upto 60m bgl. In the southern parts of Sahara block the depth to bedrock often reaches 60m bgl or more. Deepest bedrock has been encountered in southern periphery of Sahara block (more than 80m bgl) covering however, a negligible geographic area.

Depth to bedrock (m bgl)	Block wise area coverage (sq km)																						Total Area (sq km)
	Asind		Banera		Hurda		Jahazpur		Kotri		Mandal		Mandalgarh		Raipur		Sahara		Shahpura		Suwana		
	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	Area	%age	
< 20	130.3	12.2	97.7	14.0	82.0	13.4	225.4	22.5	209.9	23.0	377.5	34.0	443.0	37.0	20.5	4.1	20.2	3.0	516.5	45.0	8.5	0.9	2,131.5
20-40	935.9	87.8	585.6	86.0	530.8	86.6	776.8	77.5	710.6	77.0	651.7	59.0	626.6	53.0	471.4	93.6	335.3	49.8	632.0	55.0	915.9	99.1	7,172.6
40-60	-	-	-	-	-	-	-	-	-	-	74.1	7.0	87.9	7.0	11.7	2.3	305.6	45.4	-	-	-	-	479.3
60-80	-	-	-	-	-	-	-	-	-	-	-	-	31.1	3.0	-	-	9.7	1.4	-	-	-	-	40.8
> 80	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2.8	0.4	-	-	-	-	2.8
Total	1,066.2	100.0	683.3	100.0	612.8	100.0	1,002.2	100.0	920.5	100.0	1,103.3	100.0	1,188.6	100.0	503.6	100.0	673.6	100.0	1,148.5	100.0	924.4	100.0	9,827.0

UNCONFINED AQUIFER

Hardrock areas

Aquifers in the district are predominantly formed in weathered, fractured and jointed rock formations occurring at shallower depths and these constitute good unconfined aquifers. The thickness of aquifers thus formed varies from less than 10m to about 80m. The central and eastern parts have very thick zones especially in Sahara block and occasionally in Jahazpur and Kotri blocks. Rest of the blocks have moderate to low thickness of aquifers in hardrock.

Unconfined aquifer Thickness (m)	Block wise area coverage (sq km)											Total Area (sq km)
	Asind	Banera	Hurda	Jahazpur	Kotri	Mandal	Mandalgarh	Raipur	Sahara	Shahpura	Suwana	
< 10	287.5	115.1	117.3	61.7	4.2	121.1	334.4	162.3	74.1	260.0	295.0	1,832.7
10-20	492.1	303.3	355.2	346.5	342.3	303.4	688.6	279.7	454.2	270.1	378.0	4,213.4
20-30	202.4	225	92.4	199	311.3	339.3	134.7	60.1	91.8	427.7	243.9	2,327.6
30-40	83.0	37.1	44.9	187.6	192.1	255.9	26.8	1.5	28.4	156.3	7.5	1,021.1
40-50	1.2	2.8	3.0	121.5	46.7	83.6	4.1	-	14.8	34.4	-	312.1
50-60	-	-	-	67.3	20.6	-	-	-	5.8	-	-	93.7
60-70	-	-	-	18.6	3.3	-	-	-	3.4	-	-	25.3
> 70	-	-	-	-	-	-	-	-	1.1	-	-	1.1
Total	1,066.2	683.3	612.8	1,002.2	920.5	1,103.3	1,188.6	503.6	673.6	1,148.5	924.4	9,827.0



PLATE - XVI

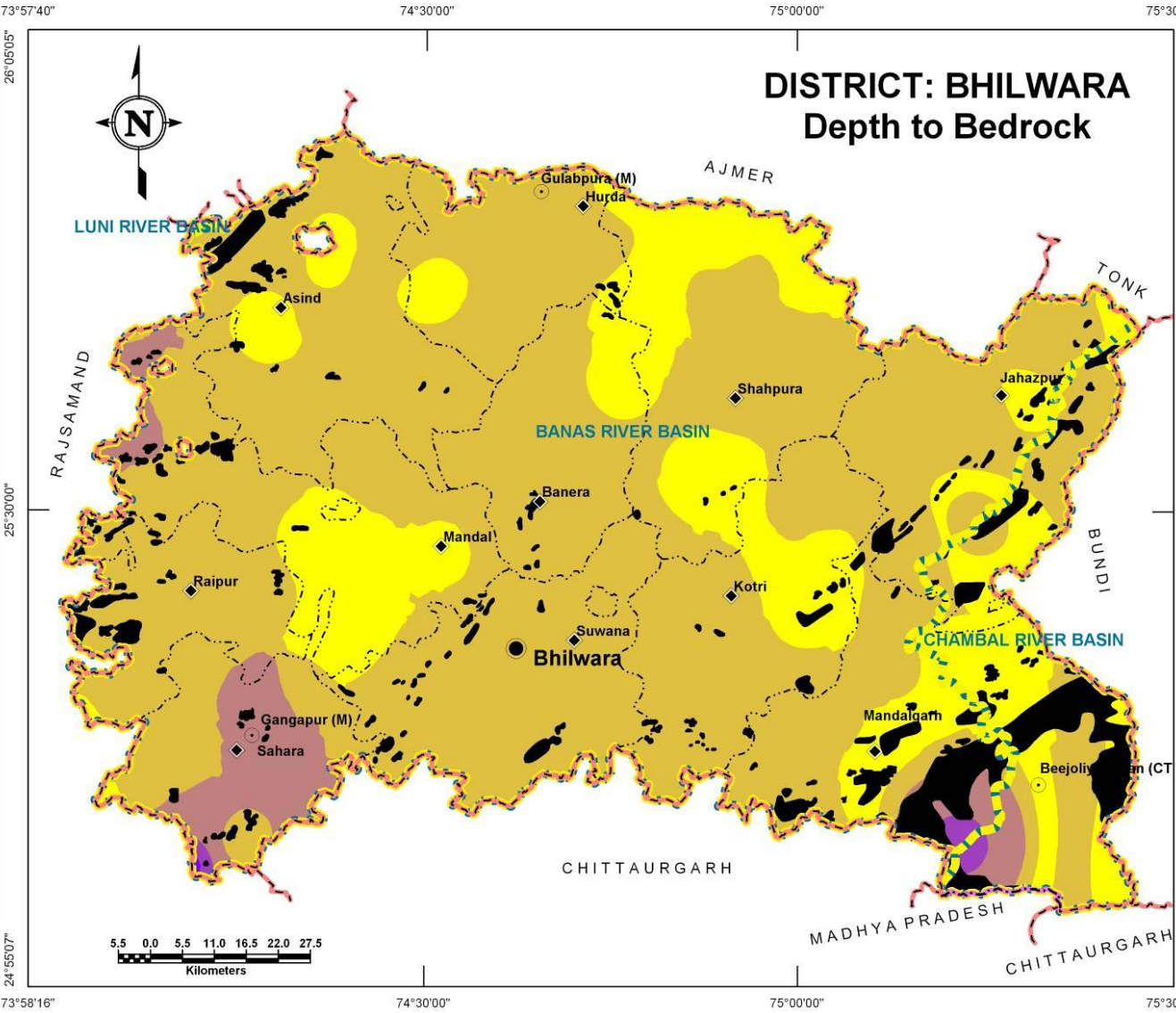
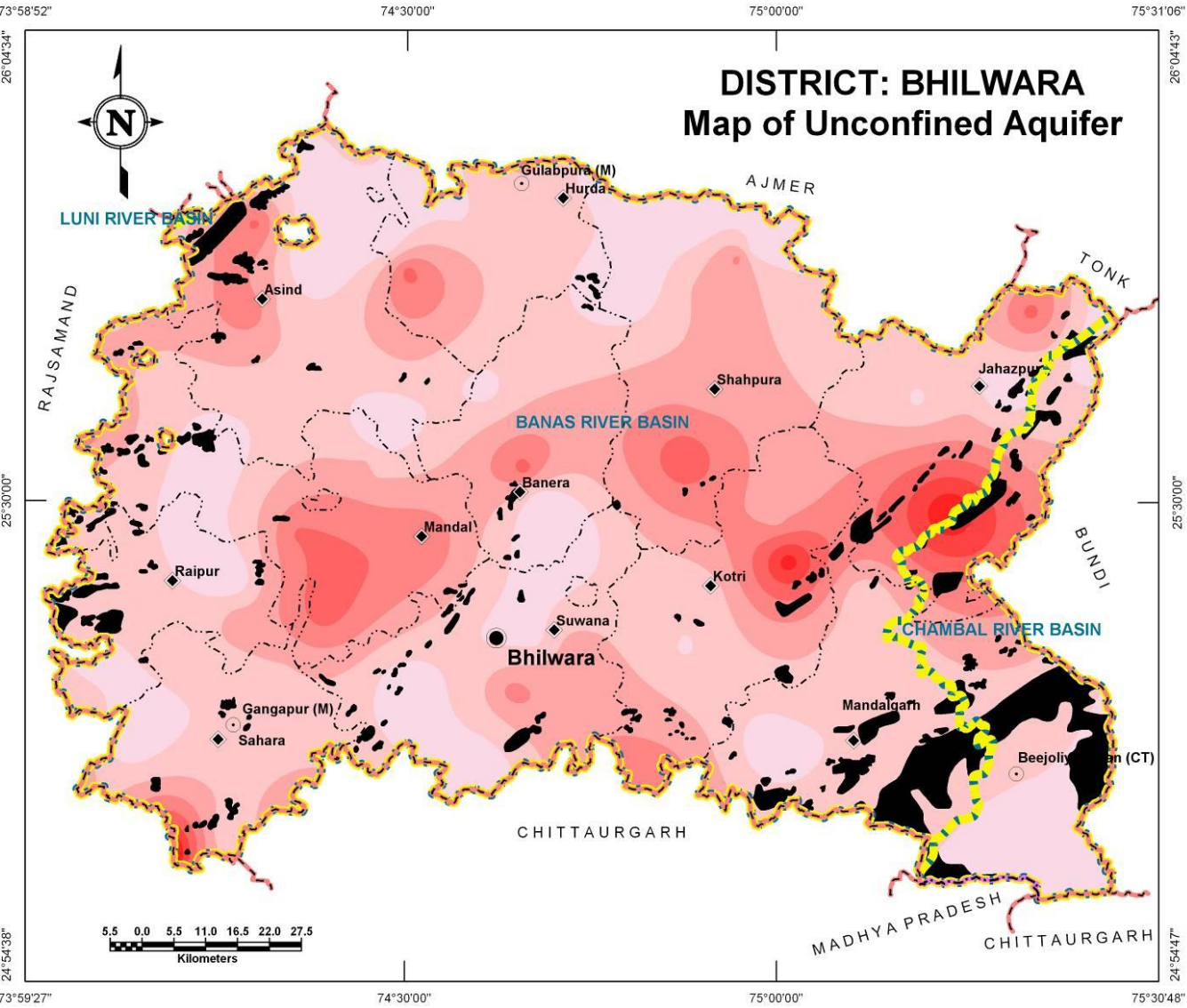


PLATE - XVII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Hills

Depth to Bedrock (m bgl):

- < 20
- 20-40
- 40-60
- 60-80
- > 80

LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary
- Hills

Aquifer Thickness(m):

- Weathered & Fractured Zone in Hardrock Area**
- < 10
- 10-20
- 20-30
- 30-40
- 40-50
- 50-60
- 60-70
- > 70

Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

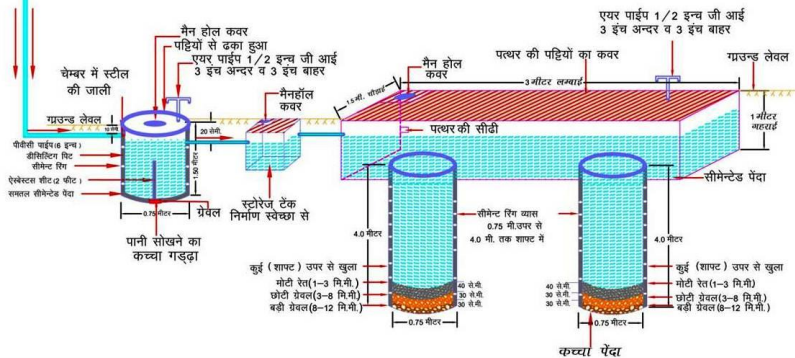
S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



भवन छत क्षेत्रफल 300 से 500 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

चित्र-4

- PVC पाईप 6" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.50 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 3 मी लम्बा x 1 मी गहरा
- सीमेन्ट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा) (संख्या 2)
- संरचना की अनुमानित लागत रु 24,000 अधिक
- वार्षिक पुनर्भरित जल लगभग 2,00,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 40,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव

बोरोन-स्नायु तन्त्र पर प्रभाव

फ्लोराइड - दंत क्षरण

क्लोराइड-सोडियम के साथ मिलकर उच्च रक्त चाप

सोडियम-हृदय, गुर्दा व रक्त परिसंचरण रोगों से ग्रसित लोगों को हानिकारक

कैल्शियम-जोड़ों में कड़ापन

नाइट्रेट-नवजात शिशुओं में ब्लू बेबी बीमारी (मेथेमोग्लोबिनमिया)

आर्सेनिक-त्वचा रोग, कैंसर

सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

लेड-बच्चों के शारीरिक व मानसिक विकास में बाधा वयस्कों में गुर्दे के रोग

आयरन-आयरन जीवाणु से आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न, हड्डियों में मुड़ाव



केन्द्रीय भूमि जल बोर्ड,
पश्चिमी क्षेत्र, जयपुर
जल संसाधन मंत्रालय
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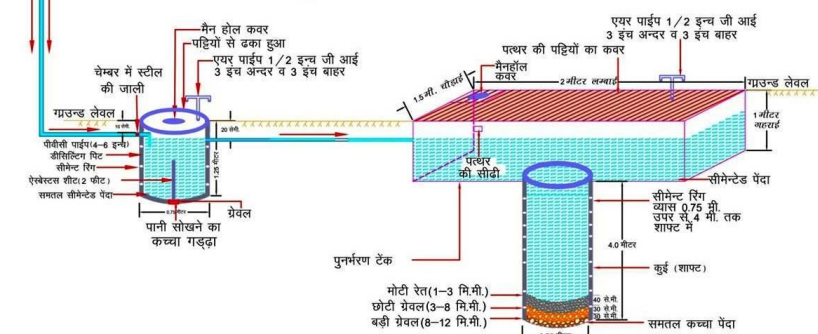
भूजल अमूल्य है इसे प्रदूषित न करें।



भवन छत क्षेत्रफल 200 से 300 वर्गमीटर तक
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

चित्र-3

- PVC पाईप 4" - 6" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 2 मी लम्बा x 1 मी गहरा
- सीमेन्ट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 15,000-16,000
- वार्षिक पुनर्भरित जल लगभग 1,25,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 25,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम

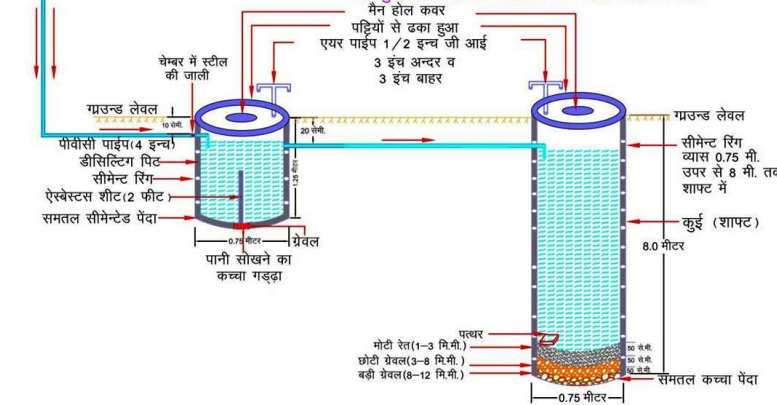


भवन छत क्षेत्रफल 100 से 200 वर्गमीटर तक

चित्र-2

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाईप 4" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेन्ट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 11,000-12,000
- वार्षिक पुनर्भरित जल लगभग 83,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 16,64,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम

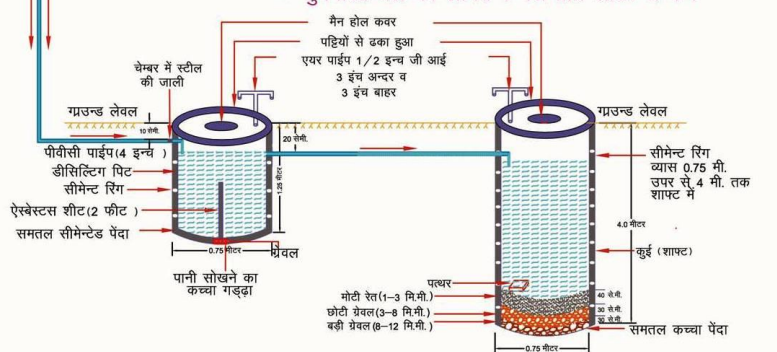


भवन छत क्षेत्रफल 100 वर्गमीटर तक

चित्र-1

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाईप 4" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेन्ट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 7000-8000
- वार्षिक पुनर्भरित जल लगभग 40,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 8,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



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